In [18]:
```
# ============================================================
# GeoAI + UNet (ResNet34, ImageNet) label-efficiency experiment
# Fractions: {10, 20, 30, 50, 100} of the train set
# Adds model complexity stats per fraction.
# ============================================================

import os, time, glob, shutil, random
from pathlib import Path
import numpy as np
from PIL import Image
import pandas as pd
import matplotlib.pyplot as plt
import torch

import geoai
import segmentation_models_pytorch as smp  # used for param counts (same arch/en

# ----------------------- setup & download -----------------------
SEED = 42
random.seed(SEED); np.random.seed(SEED); torch.manual_seed(SEED)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Using device:", device)

URL = "https://huggingface.co/datasets/giswqs/geospatial/resolve/main/waterbody-
BASE = Path(geoai.download_file(URL))  # e.g., ".../waterbody-dataset"
print("Dataset root:", BASE)

IMG_DIR = BASE / "images"
MSK_DIR = BASE / "masks"
assert IMG_DIR.exists() and MSK_DIR.exists(), "images/ or masks/ folder missing.
```

```
Using device: cuda
File already exists: waterbody-dataset.zip
Dataset root: waterbody-dataset
```

In [19]:
```
# ----------------------- utility helpers -----------------------
def list_files(folder, exts=("jpg","jpeg","png","tif","tiff")):
    files=[]
    for e in exts: files += glob.glob(str(Path(folder)/f"*.{e}"))
    return sorted(files)

def _bin_mask(path):
    # prediction/gt may be {0,1} or {0,255}; make 0/1
    a = np.array(Image.open(path).convert("L"), dtype=np.uint8)
    return (a >= 128).astype(np.uint8)
    return sorted(files)

def pair_images_masks(img_dir, msk_dir):
    imgs = list_files(img_dir)
    mask_map = {Path(p).stem: p for p in list_files(msk_dir)}
    pairs = [(i, mask_map[Path(i).stem]) for i in imgs if Path(i).stem in mask_m
    return pairs

def safe_copy(src, dst):
    dst.parent.mkdir(parents=True, exist_ok=True)
    shutil.copy2(src, dst)

def write_pairs(pairs, out_img_dir, out_msk_dir):
    out_img_dir.mkdir(parents=True, exist_ok=True)
```

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    out_msk_dir.mkdir(parents=True, exist_ok=True)
    for ip, mp in pairs:
        safe_copy(ip, out_img_dir/Path(ip).name)
        safe_copy(mp, out_msk_dir/Path(mp).name)

def load_mask_binary(path):
    # masks are 0/255 jpg/png; convert to {0,1}
    m = Image.open(path).convert("L")
    a = np.array(m, dtype=np.uint8)
    return (a >= 128).astype(np.uint8)

def overall_acc(pred, gt):
    return float((pred == gt).sum() / gt.size)

def iou_binary(pred, gt):
    inter = np.logical_and(pred==1, gt==1).sum()
    union = np.logical_or (pred==1, gt==1).sum()
    return float(inter / max(1, union))

def dice_binary(pred, gt):
    inter = np.logical_and(pred==1, gt==1).sum()
    return float((2*inter) / max(1, (pred==1).sum() + (gt==1).sum()))

def evaluate_folder(gt_dir, pred_dir, verbose=True):
    """Match GT 'water_body_XXXX' with predictions that may be named
       'water_body_XXXX.png' or 'water_body_XXXX_mask.png' (GeoAI default)."""
    gt_paths   = list_files(gt_dir)
    pred_paths = list_files(pred_dir)

    pred_map = {}
    for p in pred_paths:
        stem = Path(p).stem
        pred_map[stem] = p
        # also register without the trailing '_mask'
        if stem.endswith("_mask"):
            pred_map[stem[:-5]] = p

    n_pairs = 0
    oa_list, iou_list, dice_list = [], [], []
    for g in gt_paths:
        key = Path(g).stem
        if key not in pred_map:
            continue
        gt = _bin_mask(g)
        pr = _bin_mask(pred_map[key])
        n_pairs += 1
        # metrics
        oa   = (pr == gt).mean()
        inter = np.logical_and(pr==1, gt==1).sum()
        union = np.logical_or (pr==1, gt==1).sum()
        iou  = inter / max(1, union)
        dice = (2*inter) / max(1, (pr==1).sum() + (gt==1).sum())
        oa_list.append(oa); iou_list.append(iou); dice_list.append(dice)

    if verbose:
        print(f"[eval] matched {n_pairs}/{len(gt_paths)} files in {pred_dir}")

    if n_pairs == 0:
        return float("nan"), float("nan"), float("nan")
    return float(np.mean(oa_list)), float(np.mean(iou_list)), float(np.mean(dice
```

In [20]:
```
# ----------------------- fixed split (80/10/10) -----------------------
ALL = pair_images_masks(IMG_DIR, MSK_DIR)
assert len(ALL) > 0, "No image/mask pairs found."
rng = np.random.default_rng(SEED)
perm = rng.permutation(len(ALL))
ALL = [ALL[i] for i in perm]

n = len(ALL)
n_tr = int(round(0.8*n))
n_va = int(round(0.1*n))
train_pairs = ALL[:n_tr]
val_pairs   = ALL[n_tr:n_tr+n_va]
test_pairs  = ALL[n_tr+n_va:]

SPLIT_DIR = BASE / "fixed_split"
(TR_IMG, TR_MSK) = (SPLIT_DIR/"train_images", SPLIT_DIR/"train_masks")
(VA_IMG, VA_MSK) = (SPLIT_DIR/"val_images",   SPLIT_DIR/"val_masks")
(TE_IMG, TE_MSK) = (SPLIT_DIR/"test_images",  SPLIT_DIR/"test_masks")

# write (idempotent)
if not TR_IMG.exists():
    write_pairs(train_pairs, TR_IMG, TR_MSK)
    write_pairs(val_pairs,   VA_IMG, VA_MSK)
    write_pairs(test_pairs,  TE_IMG, TE_MSK)

print(f"Split sizes -> train:{len(list_files(TR_IMG))}  val:{len(list_files(VA_I
```

```
Split sizes -> train:2273  val:284  test:284
```

In [21]:
```
# ----------------------- build fraction subsets -----------------------
FRACTIONS = [10, 20, 30, 50, 100]  # percent
FR_BASE = BASE / "fractions"
FR_BASE.mkdir(exist_ok=True)

train_img_set = list_files(TR_IMG)
train_msk_map = {Path(p).stem: p for p in list_files(TR_MSK)}

for pct in FRACTIONS:
    out_i = FR_BASE / f"frac_{pct}" / "images"
    out_m = FR_BASE / f"frac_{pct}" / "masks"
    if out_i.exists() and len(list_files(out_i))>0:
        continue
    k = max(1, int(round(len(train_img_set)*pct/100.0)))
    sel = rng.choice(train_img_set, size=k, replace=False)
    pairs = [(p, train_msk_map[Path(p).stem]) for p in sel if Path(p).stem in tr
    write_pairs(pairs, out_i, out_m)
    print(f"Prepared fraction {pct}% -> {len(pairs)} images")
```

In [22]:
```
# ----------------------- constants (model/training) -----------------------
ARCH       = "unet"
ENCODER    = "resnet34"
ENC_WTS    = "imagenet"
IMG_SZ     = (512, 512)
EPOCHS     = 10
BATCH_SIZE = 16

# ----------------------- helpers: model stats -----------------------
def build_unet_resnet34(in_ch=3, classes=2):
    # same arch as geoai.train_segmentation_model call
```

```
    return smp.Unet(encoder_name=ENCODER, encoder_weights=ENC_WTS,
                    in_channels=in_ch, classes=classes)

def count_params(model):
    t = sum(p.numel() for p in model.parameters())
    tr = sum(p.numel() for p in model.parameters() if p.requires_grad)
    return int(t), int(tr)

# ----------------------- training & evaluation loop -----------------------
results = []

for pct in FRACTIONS:
    print(f"\n=== Fraction {pct}% ===")
    frac_img = FR_BASE / f"frac_{pct}" / "images"
    frac_msk = FR_BASE / f"frac_{pct}" / "masks"
    n_sub    = len(list_files(frac_img))
    print(f"Training subset size: {n_sub}")

    # ---- model complexity (param counts) from SMP UNet config
    tmp_model = build_unet_resnet34(3, 2)
    params_total, params_trainable = count_params(tmp_model)
    del tmp_model

    # ---- GeoAI requires a non-zero val_split to checkpoint "best model"
    internal_val_split = max(1.0 / max(1,n_sub), 0.004)  # at least ~1 image
    print(f"Internal val_split used: {internal_val_split:.4f}")

    out_models = BASE / f"unet_resnet34_frac_{pct}_models"
    out_models.mkdir(parents=True, exist_ok=True)

    # ---- Train
    t0 = time.time()
    geoai.train_segmentation_model(
        images_dir=str(frac_img),
        labels_dir=str(frac_msk),
        output_dir=str(out_models),
        architecture=ARCH,
        encoder_name=ENCODER,
        encoder_weights=ENC_WTS,
        num_channels=3,
        num_classes=2,
        batch_size=BATCH_SIZE,
        num_epochs=EPOCHS,
        learning_rate=1e-3,
        val_split=internal_val_split,    # <<< key (must be > 0)
        target_size=IMG_SZ,
        seed=SEED,
        verbose=True,
    )
    train_time = time.time() - t0
    best_model = out_models / "best_model.pth"

    # ---- Model size (saved weights on disk)
    try:
        model_size_MB = os.path.getsize(best_model) / (1024*1024)
    except FileNotFoundError:
        model_size_MB = float("nan")

    # ---- Predict on fixed external VAL (not timed)
    val_pred_dir = BASE / f"pred_val_frac_{pct}"
```

```
    val_pred_dir.mkdir(exist_ok=True)
    geoai.semantic_segmentation_batch(
        input_dir=str(VA_IMG),
        output_dir=str(val_pred_dir),
        model_path=str(best_model),
        architecture=ARCH,
        encoder_name=ENCODER,
        num_channels=3,
        num_classes=2,
        window_size=IMG_SZ[0],
        overlap=IMG_SZ[0]//2,
        batch_size=32,
        quiet=True,
    )
    val_OA, val_IoU, val_Dice = evaluate_folder(VA_MSK, val_pred_dir)

    # ---- Predict on fixed external TEST (timed to get ms/img)
    test_pred_dir = BASE / f"pred_test_frac_{pct}"
    test_pred_dir.mkdir(exist_ok=True)
    n_test_imgs = len(list_files(TE_IMG))
    t1 = time.time()
    geoai.semantic_segmentation_batch(
        input_dir=str(TE_IMG),
        output_dir=str(test_pred_dir),
        model_path=str(best_model),
        architecture=ARCH,
        encoder_name=ENCODER,
        num_channels=3,
        num_classes=2,
        window_size=IMG_SZ[0],
        overlap=IMG_SZ[0]//2,
        batch_size=32,
        quiet=True,
    )
    infer_time_s = time.time() - t1
    infer_ms_per_img = (infer_time_s / max(1,n_test_imgs)) * 1000.0

    test_OA, test_IoU, test_Dice = evaluate_folder(TE_MSK, test_pred_dir)

    row = dict(
        fraction_pct=pct,
        n_train_imgs=n_sub,
        params_total=params_total,
        params_trainable=params_trainable,
        model_size_MB=round(model_size_MB,3),
        train_time_s=round(train_time,3),
        infer_ms_per_img=round(infer_ms_per_img,3),
        val_OA=round(val_OA,4),   val_IoU=round(val_IoU,4),   val_Dice=round(val
        test_OA=round(test_OA,4), test_IoU=round(test_IoU,4), test_Dice=round(te
        ckpt=str(best_model),
    )
    results.append(row)
    print(f"VAL  -> OA:{row['val_OA']:.4f} | IoU:{row['val_IoU']:.4f} | Dice:{ro
    print(f"TEST -> OA:{row['test_OA']:.4f} | IoU:{row['test_IoU']:.4f} | Dice:{
    print(f"Model params: total={params_total:,} | trainable={params_trainable:,
    print(f"Train time: {train_time:.1f}s | Inference: {infer_ms_per_img:.1f} ms
```

```
=== Fraction 10% ===
Training subset size: 227
Internal val_split used: 0.0044
Using device: cuda
Found 227 image files and 227 label files
Training on 226 images, validating on 1 images
Testing data loader...
Data loader test passed.
Starting training with unet + resnet34
Model parameters: 24,436,514
Epoch: 1, Batch: 1/15, Loss: 0.7610, Time: 1.15s
Epoch: 1, Batch: 11/15, Loss: 0.5978, Time: 6.15s
Epoch 1/10: Train Loss: 0.5685, Val Loss: 0.9156, Val IoU: 0.3221, Val Dice: 0.43
36
Saving best model with IoU: 0.3221
Epoch: 2, Batch: 1/15, Loss: 0.6030, Time: 0.93s
Epoch: 2, Batch: 11/15, Loss: 0.4960, Time: 5.93s
Epoch 2/10: Train Loss: 0.5111, Val Loss: 26.7044, Val IoU: 0.2068, Val Dice: 0.3
021
Epoch: 3, Batch: 1/15, Loss: 0.6119, Time: 0.35s
Epoch: 3, Batch: 11/15, Loss: 0.5135, Time: 5.47s
Epoch 3/10: Train Loss: 0.5076, Val Loss: 2.1272, Val IoU: 0.2012, Val Dice: 0.29
03
Epoch: 4, Batch: 1/15, Loss: 0.4591, Time: 0.79s
Epoch: 4, Batch: 11/15, Loss: 0.4965, Time: 5.80s
Epoch 4/10: Train Loss: 0.5128, Val Loss: 0.8084, Val IoU: 0.3028, Val Dice: 0.46
38
Epoch: 5, Batch: 1/15, Loss: 0.3793, Time: 0.96s
Epoch: 5, Batch: 11/15, Loss: 0.4456, Time: 5.33s
Epoch 5/10: Train Loss: 0.5295, Val Loss: 1.0320, Val IoU: 0.2538, Val Dice: 0.39
02
Epoch: 6, Batch: 1/15, Loss: 0.5170, Time: 0.52s
Epoch: 6, Batch: 11/15, Loss: 0.4377, Time: 5.81s
Epoch 6/10: Train Loss: 0.5081, Val Loss: 0.7176, Val IoU: 0.3634, Val Dice: 0.53
30
Saving best model with IoU: 0.3634
Epoch: 7, Batch: 1/15, Loss: 0.4896, Time: 0.45s
Epoch: 7, Batch: 11/15, Loss: 0.4825, Time: 6.49s
Epoch 7/10: Train Loss: 0.4960, Val Loss: 0.7490, Val IoU: 0.3060, Val Dice: 0.46
71
Epoch: 8, Batch: 1/15, Loss: 0.4971, Time: 0.53s
Epoch: 8, Batch: 11/15, Loss: 0.4200, Time: 5.82s
Epoch 8/10: Train Loss: 0.4975, Val Loss: 0.8109, Val IoU: 0.3657, Val Dice: 0.48
25
Saving best model with IoU: 0.3657
Epoch: 9, Batch: 1/15, Loss: 0.4144, Time: 0.44s
Epoch: 9, Batch: 11/15, Loss: 0.5329, Time: 5.64s
Epoch 9/10: Train Loss: 0.5033, Val Loss: 0.5877, Val IoU: 0.4770, Val Dice: 0.62
75
Saving best model with IoU: 0.4770
Epoch: 10, Batch: 1/15, Loss: 0.5059, Time: 0.40s
Epoch: 10, Batch: 11/15, Loss: 0.4618, Time: 6.11s
Epoch 10/10: Train Loss: 0.4825, Val Loss: 0.6640, Val IoU: 0.3724, Val Dice: 0.4
916
Training complete! Best IoU: 0.4770
Models saved to waterbody-dataset\unet_resnet34_frac_10_models
Found 284 image files to process
Processing file 1/284: water_body_1044.jpg
Processing file 2/284: water_body_1064.jpg
Processing file 3/284: water_body_107.jpg
```

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Processing file 4/284: water_body_1070.jpg
Processing file 5/284: water_body_11.jpg
Processing file 6/284: water_body_1133.jpg
Processing file 7/284: water_body_1143.jpg
Processing file 8/284: water_body_1152.jpg
Processing file 9/284: water_body_1170.jpg
Processing file 10/284: water_body_118.jpg
Processing file 11/284: water_body_1195.jpg
Processing file 12/284: water_body_120.jpg
Processing file 13/284: water_body_1201.jpg
Processing file 14/284: water_body_1204.jpg
Processing file 15/284: water_body_1228.jpg
Processing file 16/284: water_body_1235.jpg
Processing file 17/284: water_body_1237.jpg
Processing file 18/284: water_body_1253.jpg
Processing file 19/284: water_body_1260.jpg
Processing file 20/284: water_body_1313.jpg
Processing file 21/284: water_body_1329.jpg
Processing file 22/284: water_body_1334.jpg
Processing file 23/284: water_body_1343.jpg
Processing file 24/284: water_body_1347.jpg
Processing file 25/284: water_body_1369.jpg
Processing file 26/284: water_body_1383.jpg
Processing file 27/284: water_body_1417.jpg
Processing file 28/284: water_body_1422.jpg
Processing file 29/284: water_body_1447.jpg
Processing file 30/284: water_body_1453.jpg
Processing file 31/284: water_body_1459.jpg
Processing file 32/284: water_body_1462.jpg
Processing file 33/284: water_body_1470.jpg
Processing file 34/284: water_body_1482.jpg
Processing file 35/284: water_body_1498.jpg
Processing file 36/284: water_body_1509.jpg
Processing file 37/284: water_body_1510.jpg
Processing file 38/284: water_body_1511.jpg
Processing file 39/284: water_body_1529.jpg
Processing file 40/284: water_body_1532.jpg
Processing file 41/284: water_body_1553.jpg
Processing file 42/284: water_body_1575.jpg
Processing file 43/284: water_body_1582.jpg
Processing file 44/284: water_body_1594.jpg
Processing file 45/284: water_body_1597.jpg
Processing file 46/284: water_body_1611.jpg
Processing file 47/284: water_body_1615.jpg
Processing file 48/284: water_body_1618.jpg
Processing file 49/284: water_body_1620.jpg
Processing file 50/284: water_body_1628.jpg
Processing file 51/284: water_body_1634.jpg
Processing file 52/284: water_body_1645.jpg
Processing file 53/284: water_body_1656.jpg
Processing file 54/284: water_body_1657.jpg
Processing file 55/284: water_body_1674.jpg
Processing file 56/284: water_body_1688.jpg
Processing file 57/284: water_body_1691.jpg
Processing file 58/284: water_body_1694.jpg
Processing file 59/284: water_body_1699.jpg
Processing file 60/284: water_body_1703.jpg
Processing file 61/284: water_body_1704.jpg
Processing file 62/284: water_body_1711.jpg
Processing file 63/284: water_body_1724.jpg
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Processing file 64/284: water_body_1734.jpg
Processing file 65/284: water_body_1745.jpg
Processing file 66/284: water_body_1749.jpg
Processing file 67/284: water_body_1770.jpg
Processing file 68/284: water_body_1771.jpg
Processing file 69/284: water_body_1776.jpg
Processing file 70/284: water_body_1794.jpg
Processing file 71/284: water_body_1816.jpg
Processing file 72/284: water_body_1832.jpg
Processing file 73/284: water_body_1833.jpg
Processing file 74/284: water_body_1839.jpg
Processing file 75/284: water_body_186.jpg
Processing file 76/284: water_body_2182.jpg
Processing file 77/284: water_body_2203.jpg
Processing file 78/284: water_body_2220.jpg
Processing file 79/284: water_body_2227.jpg
Processing file 80/284: water_body_2235.jpg
Processing file 81/284: water_body_2246.jpg
Processing file 82/284: water_body_2250.jpg
Processing file 83/284: water_body_2253.jpg
Processing file 84/284: water_body_2276.jpg
Processing file 85/284: water_body_2287.jpg
Processing file 86/284: water_body_2291.jpg
Processing file 87/284: water_body_2340.jpg
Processing file 88/284: water_body_2370.jpg
Processing file 89/284: water_body_2372.jpg
Processing file 90/284: water_body_2373.jpg
Processing file 91/284: water_body_2395.jpg
Processing file 92/284: water_body_2397.jpg
Processing file 93/284: water_body_2415.jpg
Processing file 94/284: water_body_2432.jpg
Processing file 95/284: water_body_2440.jpg
Processing file 96/284: water_body_2456.jpg
Processing file 97/284: water_body_2500.jpg
Processing file 98/284: water_body_2504.jpg
Processing file 99/284: water_body_2527.jpg
Processing file 100/284: water_body_2530.jpg
Processing file 101/284: water_body_2532.jpg
Processing file 102/284: water_body_2533.jpg
Processing file 103/284: water_body_2554.jpg
Processing file 104/284: water_body_2561.jpg
Processing file 105/284: water_body_29.jpg
Processing file 106/284: water_body_42.jpg
Processing file 107/284: water_body_59.jpg
Processing file 108/284: water_body_606.jpg
Processing file 109/284: water_body_631.jpg
Processing file 110/284: water_body_6378.jpg
Processing file 111/284: water_body_6389.jpg
Processing file 112/284: water_body_6421.jpg
Processing file 113/284: water_body_644.jpg
Processing file 114/284: water_body_65.jpg
Processing file 115/284: water_body_654.jpg
Processing file 116/284: water_body_7037.jpg
Processing file 117/284: water_body_7045.jpg
Processing file 118/284: water_body_7056.jpg
Processing file 119/284: water_body_7058.jpg
Processing file 120/284: water_body_7071.jpg
Processing file 121/284: water_body_7090.jpg
Processing file 122/284: water_body_7095.jpg
Processing file 123/284: water_body_71.jpg
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Processing file 124/284: water_body_7137.jpg
Processing file 125/284: water_body_7140.jpg
Processing file 126/284: water_body_7147.jpg
Processing file 127/284: water_body_7162.jpg
Processing file 128/284: water_body_7165.jpg
Processing file 129/284: water_body_7172.jpg
Processing file 130/284: water_body_718.jpg
Processing file 131/284: water_body_7184.jpg
Processing file 132/284: water_body_7192.jpg
Processing file 133/284: water_body_720.jpg
Processing file 134/284: water_body_7205.jpg
Processing file 135/284: water_body_7228.jpg
Processing file 136/284: water_body_7230.jpg
Processing file 137/284: water_body_7232.jpg
Processing file 138/284: water_body_7244.jpg
Processing file 139/284: water_body_7258.jpg
Processing file 140/284: water_body_7260.jpg
Processing file 141/284: water_body_7262.jpg
Processing file 142/284: water_body_7283.jpg
Processing file 143/284: water_body_7319.jpg
Processing file 144/284: water_body_7333.jpg
Processing file 145/284: water_body_7352.jpg
Processing file 146/284: water_body_7404.jpg
Processing file 147/284: water_body_7420.jpg
Processing file 148/284: water_body_7429.jpg
Processing file 149/284: water_body_7432.jpg
Processing file 150/284: water_body_7436.jpg
Processing file 151/284: water_body_7437.jpg
Processing file 152/284: water_body_7444.jpg
Processing file 153/284: water_body_7475.jpg
Processing file 154/284: water_body_7508.jpg
Processing file 155/284: water_body_7515.jpg
Processing file 156/284: water_body_7517.jpg
Processing file 157/284: water_body_7518.jpg
Processing file 158/284: water_body_7525.jpg
Processing file 159/284: water_body_7540.jpg
Processing file 160/284: water_body_7544.jpg
Processing file 161/284: water_body_7569.jpg
Processing file 162/284: water_body_7626.jpg
Processing file 163/284: water_body_7686.jpg
Processing file 164/284: water_body_7693.jpg
Processing file 165/284: water_body_7696.jpg
Processing file 166/284: water_body_7703.jpg
Processing file 167/284: water_body_7707.jpg
Processing file 168/284: water_body_7708.jpg
Processing file 169/284: water_body_7709.jpg
Processing file 170/284: water_body_7711.jpg
Processing file 171/284: water_body_772.jpg
Processing file 172/284: water_body_7720.jpg
Processing file 173/284: water_body_7727.jpg
Processing file 174/284: water_body_7729.jpg
Processing file 175/284: water_body_773.jpg
Processing file 176/284: water_body_7750.jpg
Processing file 177/284: water_body_7773.jpg
Processing file 178/284: water_body_7781.jpg
Processing file 179/284: water_body_7791.jpg
Processing file 180/284: water_body_7815.jpg
Processing file 181/284: water_body_7823.jpg
Processing file 182/284: water_body_7828.jpg
Processing file 183/284: water_body_783.jpg
```

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Processing file 184/284: water_body_7848.jpg
Processing file 185/284: water_body_7868.jpg
Processing file 186/284: water_body_7869.jpg
Processing file 187/284: water_body_7892.jpg
Processing file 188/284: water_body_7903.jpg
Processing file 189/284: water_body_7944.jpg
Processing file 190/284: water_body_7946.jpg
Processing file 191/284: water_body_7947.jpg
Processing file 192/284: water_body_7954.jpg
Processing file 193/284: water_body_7955.jpg
Processing file 194/284: water_body_7957.jpg
Processing file 195/284: water_body_7978.jpg
Processing file 196/284: water_body_8002.jpg
Processing file 197/284: water_body_8013.jpg
Processing file 198/284: water_body_8017.jpg
Processing file 199/284: water_body_8027.jpg
Processing file 200/284: water_body_8059.jpg
Processing file 201/284: water_body_806.jpg
Processing file 202/284: water_body_8068.jpg
Processing file 203/284: water_body_8076.jpg
Processing file 204/284: water_body_8098.jpg
Processing file 205/284: water_body_810.jpg
Processing file 206/284: water_body_811.jpg
Processing file 207/284: water_body_8165.jpg
Processing file 208/284: water_body_8177.jpg
Processing file 209/284: water_body_818.jpg
Processing file 210/284: water_body_8185.jpg
Processing file 211/284: water_body_8219.jpg
Processing file 212/284: water_body_823.jpg
Processing file 213/284: water_body_8242.jpg
Processing file 214/284: water_body_8248.jpg
Processing file 215/284: water_body_8253.jpg
Processing file 216/284: water_body_8319.jpg
Processing file 217/284: water_body_8338.jpg
Processing file 218/284: water_body_8344.jpg
Processing file 219/284: water_body_8348.jpg
Processing file 220/284: water_body_8356.jpg
Processing file 221/284: water_body_8377.jpg
Processing file 222/284: water_body_8387.jpg
Processing file 223/284: water_body_839.jpg
Processing file 224/284: water_body_8398.jpg
Processing file 225/284: water_body_8408.jpg
Processing file 226/284: water_body_8412.jpg
Processing file 227/284: water_body_8423.jpg
Processing file 228/284: water_body_8424.jpg
Processing file 229/284: water_body_8439.jpg
Processing file 230/284: water_body_8442.jpg
Processing file 231/284: water_body_8455.jpg
Processing file 232/284: water_body_8462.jpg
Processing file 233/284: water_body_847.jpg
Processing file 234/284: water_body_8476.jpg
Processing file 235/284: water_body_8497.jpg
Processing file 236/284: water_body_8558.jpg
Processing file 237/284: water_body_8560.jpg
Processing file 238/284: water_body_8583.jpg
Processing file 239/284: water_body_8607.jpg
Processing file 240/284: water_body_8619.jpg
Processing file 241/284: water_body_8622.jpg
Processing file 242/284: water_body_8633.jpg
Processing file 243/284: water_body_8650.jpg
```

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Processing file 244/284: water_body_8651.jpg
Processing file 245/284: water_body_8653.jpg
Processing file 246/284: water_body_8664.jpg
Processing file 247/284: water_body_8670.jpg
Processing file 248/284: water_body_8671.jpg
Processing file 249/284: water_body_8680.jpg
Processing file 250/284: water_body_8687.jpg
Processing file 251/284: water_body_8693.jpg
Processing file 252/284: water_body_8702.jpg
Processing file 253/284: water_body_8703.jpg
Processing file 254/284: water_body_8715.jpg
Processing file 255/284: water_body_8722.jpg
Processing file 256/284: water_body_8729.jpg
Processing file 257/284: water_body_875.jpg
Processing file 258/284: water_body_8760.jpg
Processing file 259/284: water_body_8763.jpg
Processing file 260/284: water_body_8764.jpg
Processing file 261/284: water_body_8789.jpg
Processing file 262/284: water_body_8823.jpg
Processing file 263/284: water_body_8825.jpg
Processing file 264/284: water_body_8843.jpg
Processing file 265/284: water_body_885.jpg
Processing file 266/284: water_body_8854.jpg
Processing file 267/284: water_body_8877.jpg
Processing file 268/284: water_body_8885.jpg
Processing file 269/284: water_body_8911.jpg
Processing file 270/284: water_body_8918.jpg
Processing file 271/284: water_body_8930.jpg
Processing file 272/284: water_body_8949.jpg
Processing file 273/284: water_body_8953.jpg
Processing file 274/284: water_body_8960.jpg
Processing file 275/284: water_body_897.jpg
Processing file 276/284: water_body_900.jpg
Processing file 277/284: water_body_938.jpg
Processing file 278/284: water_body_957.jpg
Processing file 279/284: water_body_967.jpg
Processing file 280/284: water_body_970.jpg
Processing file 281/284: water_body_978.jpg
Processing file 282/284: water_body_979.jpg
Processing file 283/284: water_body_987.jpg
Processing file 284/284: water_body_99.jpg
Batch processing completed. Results saved to waterbody-dataset\pred_val_frac_10
[eval] matched 284/284 files in waterbody-dataset\pred_val_frac_10
Found 284 image files to process
Processing file 1/284: water_body_1003.jpg
Processing file 2/284: water_body_1015.jpg
Processing file 3/284: water_body_105.jpg
Processing file 4/284: water_body_1052.jpg
Processing file 5/284: water_body_1059.jpg
Processing file 6/284: water_body_1062.jpg
Processing file 7/284: water_body_1074.jpg
Processing file 8/284: water_body_1076.jpg
Processing file 9/284: water_body_1089.jpg
Processing file 10/284: water_body_1096.jpg
Processing file 11/284: water_body_1104.jpg
Processing file 12/284: water_body_1109.jpg
Processing file 13/284: water_body_1112.jpg
Processing file 14/284: water_body_1124.jpg
Processing file 15/284: water_body_1184.jpg
Processing file 16/284: water_body_1203.jpg
```

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Processing file 17/284: water_body_121.jpg
Processing file 18/284: water_body_1227.jpg
Processing file 19/284: water_body_1280.jpg
Processing file 20/284: water_body_1307.jpg
Processing file 21/284: water_body_1354.jpg
Processing file 22/284: water_body_1359.jpg
Processing file 23/284: water_body_1361.jpg
Processing file 24/284: water_body_1368.jpg
Processing file 25/284: water_body_1373.jpg
Processing file 26/284: water_body_1382.jpg
Processing file 27/284: water_body_1392.jpg
Processing file 28/284: water_body_1406.jpg
Processing file 29/284: water_body_1420.jpg
Processing file 30/284: water_body_1433.jpg
Processing file 31/284: water_body_1439.jpg
Processing file 32/284: water_body_1451.jpg
Processing file 33/284: water_body_1468.jpg
Processing file 34/284: water_body_1502.jpg
Processing file 35/284: water_body_1513.jpg
Processing file 36/284: water_body_1525.jpg
Processing file 37/284: water_body_1559.jpg
Processing file 38/284: water_body_1565.jpg
Processing file 39/284: water_body_1572.jpg
Processing file 40/284: water_body_1577.jpg
Processing file 41/284: water_body_1579.jpg
Processing file 42/284: water_body_1586.jpg
Processing file 43/284: water_body_1591.jpg
Processing file 44/284: water_body_1593.jpg
Processing file 45/284: water_body_1622.jpg
Processing file 46/284: water_body_1623.jpg
Processing file 47/284: water_body_1626.jpg
Processing file 48/284: water_body_1630.jpg
Processing file 49/284: water_body_1655.jpg
Processing file 50/284: water_body_1658.jpg
Processing file 51/284: water_body_1662.jpg
Processing file 52/284: water_body_1663.jpg
Processing file 53/284: water_body_1664.jpg
Processing file 54/284: water_body_1684.jpg
Processing file 55/284: water_body_1693.jpg
Processing file 56/284: water_body_1733.jpg
Processing file 57/284: water_body_1760.jpg
Processing file 58/284: water_body_1768.jpg
Processing file 59/284: water_body_1769.jpg
Processing file 60/284: water_body_1787.jpg
Processing file 61/284: water_body_1796.jpg
Processing file 62/284: water_body_1801.jpg
Processing file 63/284: water_body_1813.jpg
Processing file 64/284: water_body_1815.jpg
Processing file 65/284: water_body_1831.jpg
Processing file 66/284: water_body_1844.jpg
Processing file 67/284: water_body_1847.jpg
Processing file 68/284: water_body_21.jpg
Processing file 69/284: water_body_2171.jpg
Processing file 70/284: water_body_2185.jpg
Processing file 71/284: water_body_2190.jpg
Processing file 72/284: water_body_22.jpg
Processing file 73/284: water_body_2217.jpg
Processing file 74/284: water_body_2255.jpg
Processing file 75/284: water_body_2264.jpg
Processing file 76/284: water_body_2270.jpg
```

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Processing file 77/284: water_body_2282.jpg
Processing file 78/284: water_body_2285.jpg
Processing file 79/284: water_body_2293.jpg
Processing file 80/284: water_body_2295.jpg
Processing file 81/284: water_body_2299.jpg
Processing file 82/284: water_body_2307.jpg
Processing file 83/284: water_body_232.jpg
Processing file 84/284: water_body_2320.jpg
Processing file 85/284: water_body_2333.jpg
Processing file 86/284: water_body_2385.jpg
Processing file 87/284: water_body_2402.jpg
Processing file 88/284: water_body_2472.jpg
Processing file 89/284: water_body_2474.jpg
Processing file 90/284: water_body_2498.jpg
Processing file 91/284: water_body_2520.jpg
Processing file 92/284: water_body_2529.jpg
Processing file 93/284: water_body_2538.jpg
Processing file 94/284: water_body_2541.jpg
Processing file 95/284: water_body_2553.jpg
Processing file 96/284: water_body_2559.jpg
Processing file 97/284: water_body_27.jpg
Processing file 98/284: water_body_39.jpg
Processing file 99/284: water_body_55.jpg
Processing file 100/284: water_body_590.jpg
Processing file 101/284: water_body_599.jpg
Processing file 102/284: water_body_607.jpg
Processing file 103/284: water_body_615.jpg
Processing file 104/284: water_body_620.jpg
Processing file 105/284: water_body_6377.jpg
Processing file 106/284: water_body_6385.jpg
Processing file 107/284: water_body_64.jpg
Processing file 108/284: water_body_6426.jpg
Processing file 109/284: water_body_668.jpg
Processing file 110/284: water_body_68.jpg
Processing file 111/284: water_body_690.jpg
Processing file 112/284: water_body_693.jpg
Processing file 113/284: water_body_7029.jpg
Processing file 114/284: water_body_704.jpg
Processing file 115/284: water_body_7040.jpg
Processing file 116/284: water_body_7052.jpg
Processing file 117/284: water_body_7057.jpg
Processing file 118/284: water_body_7067.jpg
Processing file 119/284: water_body_7068.jpg
Processing file 120/284: water_body_7079.jpg
Processing file 121/284: water_body_7100.jpg
Processing file 122/284: water_body_7132.jpg
Processing file 123/284: water_body_714.jpg
Processing file 124/284: water_body_7166.jpg
Processing file 125/284: water_body_7168.jpg
Processing file 126/284: water_body_7173.jpg
Processing file 127/284: water_body_7181.jpg
Processing file 128/284: water_body_7191.jpg
Processing file 129/284: water_body_7199.jpg
Processing file 130/284: water_body_72.jpg
Processing file 131/284: water_body_7207.jpg
Processing file 132/284: water_body_7235.jpg
Processing file 133/284: water_body_7238.jpg
Processing file 134/284: water_body_7245.jpg
Processing file 135/284: water_body_7247.jpg
Processing file 136/284: water_body_7252.jpg
```

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Processing file 137/284: water_body_7265.jpg
Processing file 138/284: water_body_7275.jpg
Processing file 139/284: water_body_7276.jpg
Processing file 140/284: water_body_7278.jpg
Processing file 141/284: water_body_7289.jpg
Processing file 142/284: water_body_7290.jpg
Processing file 143/284: water_body_7296.jpg
Processing file 144/284: water_body_7305.jpg
Processing file 145/284: water_body_7327.jpg
Processing file 146/284: water_body_7386.jpg
Processing file 147/284: water_body_7392.jpg
Processing file 148/284: water_body_7395.jpg
Processing file 149/284: water_body_7396.jpg
Processing file 150/284: water_body_7401.jpg
Processing file 151/284: water_body_7413.jpg
Processing file 152/284: water_body_7416.jpg
Processing file 153/284: water_body_7417.jpg
Processing file 154/284: water_body_7418.jpg
Processing file 155/284: water_body_742.jpg
Processing file 156/284: water_body_7421.jpg
Processing file 157/284: water_body_7423.jpg
Processing file 158/284: water_body_7433.jpg
Processing file 159/284: water_body_7438.jpg
Processing file 160/284: water_body_744.jpg
Processing file 161/284: water_body_7486.jpg
Processing file 162/284: water_body_7499.jpg
Processing file 163/284: water_body_7510.jpg
Processing file 164/284: water_body_7511.jpg
Processing file 165/284: water_body_7543.jpg
Processing file 166/284: water_body_7552.jpg
Processing file 167/284: water_body_7562.jpg
Processing file 168/284: water_body_7575.jpg
Processing file 169/284: water_body_7578.jpg
Processing file 170/284: water_body_7581.jpg
Processing file 171/284: water_body_7586.jpg
Processing file 172/284: water_body_759.jpg
Processing file 173/284: water_body_7603.jpg
Processing file 174/284: water_body_7615.jpg
Processing file 175/284: water_body_7618.jpg
Processing file 176/284: water_body_7629.jpg
Processing file 177/284: water_body_7638.jpg
Processing file 178/284: water_body_7673.jpg
Processing file 179/284: water_body_7701.jpg
Processing file 180/284: water_body_771.jpg
Processing file 181/284: water_body_7712.jpg
Processing file 182/284: water_body_7717.jpg
Processing file 183/284: water_body_7766.jpg
Processing file 184/284: water_body_7793.jpg
Processing file 185/284: water_body_78.jpg
Processing file 186/284: water_body_7810.jpg
Processing file 187/284: water_body_7843.jpg
Processing file 188/284: water_body_7871.jpg
Processing file 189/284: water_body_788.jpg
Processing file 190/284: water_body_7891.jpg
Processing file 191/284: water_body_790.jpg
Processing file 192/284: water_body_7901.jpg
Processing file 193/284: water_body_7914.jpg
Processing file 194/284: water_body_7918.jpg
Processing file 195/284: water_body_7956.jpg
Processing file 196/284: water_body_796.jpg
```

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Processing file 197/284: water_body_7961.jpg
Processing file 198/284: water_body_7975.jpg
Processing file 199/284: water_body_7977.jpg
Processing file 200/284: water_body_800.jpg
Processing file 201/284: water_body_8005.jpg
Processing file 202/284: water_body_801.jpg
Processing file 203/284: water_body_8018.jpg
Processing file 204/284: water_body_8033.jpg
Processing file 205/284: water_body_8052.jpg
Processing file 206/284: water_body_8064.jpg
Processing file 207/284: water_body_8140.jpg
Processing file 208/284: water_body_8146.jpg
Processing file 209/284: water_body_8156.jpg
Processing file 210/284: water_body_8159.jpg
Processing file 211/284: water_body_8169.jpg
Processing file 212/284: water_body_8195.jpg
Processing file 213/284: water_body_8206.jpg
Processing file 214/284: water_body_8245.jpg
Processing file 215/284: water_body_825.jpg
Processing file 216/284: water_body_8254.jpg
Processing file 217/284: water_body_8273.jpg
Processing file 218/284: water_body_8288.jpg
Processing file 219/284: water_body_8321.jpg
Processing file 220/284: water_body_8352.jpg
Processing file 221/284: water_body_8367.jpg
Processing file 222/284: water_body_8376.jpg
Processing file 223/284: water_body_838.jpg
Processing file 224/284: water_body_8388.jpg
Processing file 225/284: water_body_8402.jpg
Processing file 226/284: water_body_8405.jpg
Processing file 227/284: water_body_841.jpg
Processing file 228/284: water_body_8446.jpg
Processing file 229/284: water_body_8448.jpg
Processing file 230/284: water_body_8463.jpg
Processing file 231/284: water_body_8467.jpg
Processing file 232/284: water_body_8482.jpg
Processing file 233/284: water_body_8491.jpg
Processing file 234/284: water_body_8495.jpg
Processing file 235/284: water_body_8501.jpg
Processing file 236/284: water_body_8504.jpg
Processing file 237/284: water_body_8506.jpg
Processing file 238/284: water_body_8509.jpg
Processing file 239/284: water_body_851.jpg
Processing file 240/284: water_body_8526.jpg
Processing file 241/284: water_body_8533.jpg
Processing file 242/284: water_body_8546.jpg
Processing file 243/284: water_body_8556.jpg
Processing file 244/284: water_body_8561.jpg
Processing file 245/284: water_body_8573.jpg
Processing file 246/284: water_body_8584.jpg
Processing file 247/284: water_body_8616.jpg
Processing file 248/284: water_body_8632.jpg
Processing file 249/284: water_body_8636.jpg
Processing file 250/284: water_body_8641.jpg
Processing file 251/284: water_body_8655.jpg
Processing file 252/284: water_body_8659.jpg
Processing file 253/284: water_body_8676.jpg
Processing file 254/284: water_body_8678.jpg
Processing file 255/284: water_body_8679.jpg
Processing file 256/284: water_body_8694.jpg
```

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Processing file 257/284: water_body_8717.jpg
Processing file 258/284: water_body_8728.jpg
Processing file 259/284: water_body_8730.jpg
Processing file 260/284: water_body_8738.jpg
Processing file 261/284: water_body_8749.jpg
Processing file 262/284: water_body_8752.jpg
Processing file 263/284: water_body_8753.jpg
Processing file 264/284: water_body_8770.jpg
Processing file 265/284: water_body_8774.jpg
Processing file 266/284: water_body_8787.jpg
Processing file 267/284: water_body_8792.jpg
Processing file 268/284: water_body_881.jpg
Processing file 269/284: water_body_8847.jpg
Processing file 270/284: water_body_8850.jpg
Processing file 271/284: water_body_8851.jpg
Processing file 272/284: water_body_8859.jpg
Processing file 273/284: water_body_889.jpg
Processing file 274/284: water_body_8912.jpg
Processing file 275/284: water_body_8921.jpg
Processing file 276/284: water_body_8926.jpg
Processing file 277/284: water_body_8934.jpg
Processing file 278/284: water_body_8935.jpg
Processing file 279/284: water_body_918.jpg
Processing file 280/284: water_body_949.jpg
Processing file 281/284: water_body_96.jpg
Processing file 282/284: water_body_964.jpg
Processing file 283/284: water_body_968.jpg
Processing file 284/284: water_body_994.jpg
Batch processing completed. Results saved to waterbody-dataset\pred_test_frac_10
[eval] matched 284/284 files in waterbody-dataset\pred_test_frac_10
VAL  -> OA:0.7845 | IoU:0.3056 | Dice:0.3667
TEST -> OA:0.7756 | IoU:0.3291 | Dice:0.3946
Model params: total=24,436,514 | trainable=24,436,514 | size=93.38 MB
Train time: 84.3s | Inference: 229.4 ms/img

=== Fraction 20% ===
Training subset size: 455
Internal val_split used: 0.0040
Using device: cuda
Found 455 image files and 455 label files
Training on 453 images, validating on 2 images
Testing data loader...
Data loader test passed.
Starting training with unet + resnet34
Model parameters: 24,436,514
Epoch: 1, Batch: 1/29, Loss: 0.7690, Time: 1.08s
Epoch: 1, Batch: 11/29, Loss: 0.5086, Time: 5.68s
Epoch: 1, Batch: 21/29, Loss: 0.4498, Time: 5.49s
Epoch 1/10: Train Loss: 0.5196, Val Loss: 0.3965, Val IoU: 0.5358, Val Dice: 0.64
41
Saving best model with IoU: 0.5358
Epoch: 2, Batch: 1/29, Loss: 0.5406, Time: 0.54s
Epoch: 2, Batch: 11/29, Loss: 0.3402, Time: 6.53s
Epoch: 2, Batch: 21/29, Loss: 0.5013, Time: 4.77s
Epoch 2/10: Train Loss: 0.4584, Val Loss: 0.3427, Val IoU: 0.6693, Val Dice: 0.78
74
Saving best model with IoU: 0.6693
Epoch: 3, Batch: 1/29, Loss: 0.4261, Time: 0.36s
Epoch: 3, Batch: 11/29, Loss: 0.4526, Time: 5.83s
Epoch: 3, Batch: 21/29, Loss: 0.3824, Time: 5.56s
```

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Epoch 3/10: Train Loss: 0.4390, Val Loss: 0.5626, Val IoU: 0.4716, Val Dice: 0.59
66
Epoch: 4, Batch: 1/29, Loss: 0.3755, Time: 0.43s
Epoch: 4, Batch: 11/29, Loss: 0.5205, Time: 5.81s
Epoch: 4, Batch: 21/29, Loss: 0.3435, Time: 5.81s
Epoch 4/10: Train Loss: 0.4314, Val Loss: 0.2884, Val IoU: 0.6695, Val Dice: 0.78
13
Saving best model with IoU: 0.6695
Epoch: 5, Batch: 1/29, Loss: 0.4654, Time: 0.51s
Epoch: 5, Batch: 11/29, Loss: 0.3670, Time: 5.59s
Epoch: 5, Batch: 21/29, Loss: 0.4236, Time: 5.12s
Epoch 5/10: Train Loss: 0.4120, Val Loss: 0.2807, Val IoU: 0.6834, Val Dice: 0.79
64
Saving best model with IoU: 0.6834
Epoch: 6, Batch: 1/29, Loss: 0.3276, Time: 0.39s
Epoch: 6, Batch: 11/29, Loss: 0.5940, Time: 4.99s
Epoch: 6, Batch: 21/29, Loss: 0.3863, Time: 5.64s
Epoch 6/10: Train Loss: 0.3994, Val Loss: 0.4117, Val IoU: 0.5445, Val Dice: 0.66
78
Epoch: 7, Batch: 1/29, Loss: 0.3960, Time: 0.57s
Epoch: 7, Batch: 11/29, Loss: 0.2839, Time: 5.38s
Epoch: 7, Batch: 21/29, Loss: 0.3253, Time: 5.85s
Epoch 7/10: Train Loss: 0.3958, Val Loss: 0.3455, Val IoU: 0.6933, Val Dice: 0.80
63
Saving best model with IoU: 0.6933
Epoch: 8, Batch: 1/29, Loss: 0.3484, Time: 0.42s
Epoch: 8, Batch: 11/29, Loss: 0.3608, Time: 4.85s
Epoch: 8, Batch: 21/29, Loss: 0.3389, Time: 6.23s
Epoch 8/10: Train Loss: 0.4002, Val Loss: 0.3038, Val IoU: 0.6913, Val Dice: 0.80
30
Epoch: 9, Batch: 1/29, Loss: 0.3137, Time: 0.58s
Epoch: 9, Batch: 11/29, Loss: 0.3058, Time: 5.64s
Epoch: 9, Batch: 21/29, Loss: 0.4910, Time: 4.86s
Epoch 9/10: Train Loss: 0.4156, Val Loss: 0.5604, Val IoU: 0.5586, Val Dice: 0.69
51
Epoch: 10, Batch: 1/29, Loss: 0.4119, Time: 0.32s
Epoch: 10, Batch: 11/29, Loss: 0.4330, Time: 5.77s
Epoch: 10, Batch: 21/29, Loss: 0.4837, Time: 4.80s
Epoch 10/10: Train Loss: 0.4148, Val Loss: 0.5268, Val IoU: 0.4871, Val Dice: 0.6
127
Training complete! Best IoU: 0.6933
Models saved to waterbody-dataset\unet_resnet34_frac_20_models
Found 284 image files to process
Processing file 1/284: water_body_1044.jpg
Processing file 2/284: water_body_1064.jpg
Processing file 3/284: water_body_107.jpg
Processing file 4/284: water_body_1070.jpg
Processing file 5/284: water_body_11.jpg
Processing file 6/284: water_body_1133.jpg
Processing file 7/284: water_body_1143.jpg
Processing file 8/284: water_body_1152.jpg
Processing file 9/284: water_body_1170.jpg
Processing file 10/284: water_body_118.jpg
Processing file 11/284: water_body_1195.jpg
Processing file 12/284: water_body_120.jpg
Processing file 13/284: water_body_1201.jpg
Processing file 14/284: water_body_1204.jpg
Processing file 15/284: water_body_1228.jpg
Processing file 16/284: water_body_1235.jpg
Processing file 17/284: water_body_1237.jpg
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Processing file 18/284: water_body_1253.jpg
Processing file 19/284: water_body_1260.jpg
Processing file 20/284: water_body_1313.jpg
Processing file 21/284: water_body_1329.jpg
Processing file 22/284: water_body_1334.jpg
Processing file 23/284: water_body_1343.jpg
Processing file 24/284: water_body_1347.jpg
Processing file 25/284: water_body_1369.jpg
Processing file 26/284: water_body_1383.jpg
Processing file 27/284: water_body_1417.jpg
Processing file 28/284: water_body_1422.jpg
Processing file 29/284: water_body_1447.jpg
Processing file 30/284: water_body_1453.jpg
Processing file 31/284: water_body_1459.jpg
Processing file 32/284: water_body_1462.jpg
Processing file 33/284: water_body_1470.jpg
Processing file 34/284: water_body_1482.jpg
Processing file 35/284: water_body_1498.jpg
Processing file 36/284: water_body_1509.jpg
Processing file 37/284: water_body_1510.jpg
Processing file 38/284: water_body_1511.jpg
Processing file 39/284: water_body_1529.jpg
Processing file 40/284: water_body_1532.jpg
Processing file 41/284: water_body_1553.jpg
Processing file 42/284: water_body_1575.jpg
Processing file 43/284: water_body_1582.jpg
Processing file 44/284: water_body_1594.jpg
Processing file 45/284: water_body_1597.jpg
Processing file 46/284: water_body_1611.jpg
Processing file 47/284: water_body_1615.jpg
Processing file 48/284: water_body_1618.jpg
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Processing file 51/284: water_body_1634.jpg
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Processing file 55/284: water_body_1674.jpg
Processing file 56/284: water_body_1688.jpg
Processing file 57/284: water_body_1691.jpg
Processing file 58/284: water_body_1694.jpg
Processing file 59/284: water_body_1699.jpg
Processing file 60/284: water_body_1703.jpg
Processing file 61/284: water_body_1704.jpg
Processing file 62/284: water_body_1711.jpg
Processing file 63/284: water_body_1724.jpg
Processing file 64/284: water_body_1734.jpg
Processing file 65/284: water_body_1745.jpg
Processing file 66/284: water_body_1749.jpg
Processing file 67/284: water_body_1770.jpg
Processing file 68/284: water_body_1771.jpg
Processing file 69/284: water_body_1776.jpg
Processing file 70/284: water_body_1794.jpg
Processing file 71/284: water_body_1816.jpg
Processing file 72/284: water_body_1832.jpg
Processing file 73/284: water_body_1833.jpg
Processing file 74/284: water_body_1839.jpg
Processing file 75/284: water_body_186.jpg
Processing file 76/284: water_body_2182.jpg
Processing file 77/284: water_body_2203.jpg
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Processing file 78/284: water_body_2220.jpg
Processing file 79/284: water_body_2227.jpg
Processing file 80/284: water_body_2235.jpg
Processing file 81/284: water_body_2246.jpg
Processing file 82/284: water_body_2250.jpg
Processing file 83/284: water_body_2253.jpg
Processing file 84/284: water_body_2276.jpg
Processing file 85/284: water_body_2287.jpg
Processing file 86/284: water_body_2291.jpg
Processing file 87/284: water_body_2340.jpg
Processing file 88/284: water_body_2370.jpg
Processing file 89/284: water_body_2372.jpg
Processing file 90/284: water_body_2373.jpg
Processing file 91/284: water_body_2395.jpg
Processing file 92/284: water_body_2397.jpg
Processing file 93/284: water_body_2415.jpg
Processing file 94/284: water_body_2432.jpg
Processing file 95/284: water_body_2440.jpg
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Processing file 97/284: water_body_2500.jpg
Processing file 98/284: water_body_2504.jpg
Processing file 99/284: water_body_2527.jpg
Processing file 100/284: water_body_2530.jpg
Processing file 101/284: water_body_2532.jpg
Processing file 102/284: water_body_2533.jpg
Processing file 103/284: water_body_2554.jpg
Processing file 104/284: water_body_2561.jpg
Processing file 105/284: water_body_29.jpg
Processing file 106/284: water_body_42.jpg
Processing file 107/284: water_body_59.jpg
Processing file 108/284: water_body_606.jpg
Processing file 109/284: water_body_631.jpg
Processing file 110/284: water_body_6378.jpg
Processing file 111/284: water_body_6389.jpg
Processing file 112/284: water_body_6421.jpg
Processing file 113/284: water_body_644.jpg
Processing file 114/284: water_body_65.jpg
Processing file 115/284: water_body_654.jpg
Processing file 116/284: water_body_7037.jpg
Processing file 117/284: water_body_7045.jpg
Processing file 118/284: water_body_7056.jpg
Processing file 119/284: water_body_7058.jpg
Processing file 120/284: water_body_7071.jpg
Processing file 121/284: water_body_7090.jpg
Processing file 122/284: water_body_7095.jpg
Processing file 123/284: water_body_71.jpg
Processing file 124/284: water_body_7137.jpg
Processing file 125/284: water_body_7140.jpg
Processing file 126/284: water_body_7147.jpg
Processing file 127/284: water_body_7162.jpg
Processing file 128/284: water_body_7165.jpg
Processing file 129/284: water_body_7172.jpg
Processing file 130/284: water_body_718.jpg
Processing file 131/284: water_body_7184.jpg
Processing file 132/284: water_body_7192.jpg
Processing file 133/284: water_body_720.jpg
Processing file 134/284: water_body_7205.jpg
Processing file 135/284: water_body_7228.jpg
Processing file 136/284: water_body_7230.jpg
Processing file 137/284: water_body_7232.jpg
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Processing file 138/284: water_body_7244.jpg
Processing file 139/284: water_body_7258.jpg
Processing file 140/284: water_body_7260.jpg
Processing file 141/284: water_body_7262.jpg
Processing file 142/284: water_body_7283.jpg
Processing file 143/284: water_body_7319.jpg
Processing file 144/284: water_body_7333.jpg
Processing file 145/284: water_body_7352.jpg
Processing file 146/284: water_body_7404.jpg
Processing file 147/284: water_body_7420.jpg
Processing file 148/284: water_body_7429.jpg
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Processing file 150/284: water_body_7436.jpg
Processing file 151/284: water_body_7437.jpg
Processing file 152/284: water_body_7444.jpg
Processing file 153/284: water_body_7475.jpg
Processing file 154/284: water_body_7508.jpg
Processing file 155/284: water_body_7515.jpg
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Processing file 157/284: water_body_7518.jpg
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Processing file 162/284: water_body_7626.jpg
Processing file 163/284: water_body_7686.jpg
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Processing file 166/284: water_body_7703.jpg
Processing file 167/284: water_body_7707.jpg
Processing file 168/284: water_body_7708.jpg
Processing file 169/284: water_body_7709.jpg
Processing file 170/284: water_body_7711.jpg
Processing file 171/284: water_body_772.jpg
Processing file 172/284: water_body_7720.jpg
Processing file 173/284: water_body_7727.jpg
Processing file 174/284: water_body_7729.jpg
Processing file 175/284: water_body_773.jpg
Processing file 176/284: water_body_7750.jpg
Processing file 177/284: water_body_7773.jpg
Processing file 178/284: water_body_7781.jpg
Processing file 179/284: water_body_7791.jpg
Processing file 180/284: water_body_7815.jpg
Processing file 181/284: water_body_7823.jpg
Processing file 182/284: water_body_7828.jpg
Processing file 183/284: water_body_783.jpg
Processing file 184/284: water_body_7848.jpg
Processing file 185/284: water_body_7868.jpg
Processing file 186/284: water_body_7869.jpg
Processing file 187/284: water_body_7892.jpg
Processing file 188/284: water_body_7903.jpg
Processing file 189/284: water_body_7944.jpg
Processing file 190/284: water_body_7946.jpg
Processing file 191/284: water_body_7947.jpg
Processing file 192/284: water_body_7954.jpg
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Processing file 194/284: water_body_7957.jpg
Processing file 195/284: water_body_7978.jpg
Processing file 196/284: water_body_8002.jpg
Processing file 197/284: water_body_8013.jpg
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Processing file 198/284: water_body_8017.jpg
Processing file 199/284: water_body_8027.jpg
Processing file 200/284: water_body_8059.jpg
Processing file 201/284: water_body_806.jpg
Processing file 202/284: water_body_8068.jpg
Processing file 203/284: water_body_8076.jpg
Processing file 204/284: water_body_8098.jpg
Processing file 205/284: water_body_810.jpg
Processing file 206/284: water_body_811.jpg
Processing file 207/284: water_body_8165.jpg
Processing file 208/284: water_body_8177.jpg
Processing file 209/284: water_body_818.jpg
Processing file 210/284: water_body_8185.jpg
Processing file 211/284: water_body_8219.jpg
Processing file 212/284: water_body_823.jpg
Processing file 213/284: water_body_8242.jpg
Processing file 214/284: water_body_8248.jpg
Processing file 215/284: water_body_8253.jpg
Processing file 216/284: water_body_8319.jpg
Processing file 217/284: water_body_8338.jpg
Processing file 218/284: water_body_8344.jpg
Processing file 219/284: water_body_8348.jpg
Processing file 220/284: water_body_8356.jpg
Processing file 221/284: water_body_8377.jpg
Processing file 222/284: water_body_8387.jpg
Processing file 223/284: water_body_839.jpg
Processing file 224/284: water_body_8398.jpg
Processing file 225/284: water_body_8408.jpg
Processing file 226/284: water_body_8412.jpg
Processing file 227/284: water_body_8423.jpg
Processing file 228/284: water_body_8424.jpg
Processing file 229/284: water_body_8439.jpg
Processing file 230/284: water_body_8442.jpg
Processing file 231/284: water_body_8455.jpg
Processing file 232/284: water_body_8462.jpg
Processing file 233/284: water_body_847.jpg
Processing file 234/284: water_body_8476.jpg
Processing file 235/284: water_body_8497.jpg
Processing file 236/284: water_body_8558.jpg
Processing file 237/284: water_body_8560.jpg
Processing file 238/284: water_body_8583.jpg
Processing file 239/284: water_body_8607.jpg
Processing file 240/284: water_body_8619.jpg
Processing file 241/284: water_body_8622.jpg
Processing file 242/284: water_body_8633.jpg
Processing file 243/284: water_body_8650.jpg
Processing file 244/284: water_body_8651.jpg
Processing file 245/284: water_body_8653.jpg
Processing file 246/284: water_body_8664.jpg
Processing file 247/284: water_body_8670.jpg
Processing file 248/284: water_body_8671.jpg
Processing file 249/284: water_body_8680.jpg
Processing file 250/284: water_body_8687.jpg
Processing file 251/284: water_body_8693.jpg
Processing file 252/284: water_body_8702.jpg
Processing file 253/284: water_body_8703.jpg
Processing file 254/284: water_body_8715.jpg
Processing file 255/284: water_body_8722.jpg
Processing file 256/284: water_body_8729.jpg
Processing file 257/284: water_body_875.jpg
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Processing file 258/284: water_body_8760.jpg
Processing file 259/284: water_body_8763.jpg
Processing file 260/284: water_body_8764.jpg
Processing file 261/284: water_body_8789.jpg
Processing file 262/284: water_body_8823.jpg
Processing file 263/284: water_body_8825.jpg
Processing file 264/284: water_body_8843.jpg
Processing file 265/284: water_body_885.jpg
Processing file 266/284: water_body_8854.jpg
Processing file 267/284: water_body_8877.jpg
Processing file 268/284: water_body_8885.jpg
Processing file 269/284: water_body_8911.jpg
Processing file 270/284: water_body_8918.jpg
Processing file 271/284: water_body_8930.jpg
Processing file 272/284: water_body_8949.jpg
Processing file 273/284: water_body_8953.jpg
Processing file 274/284: water_body_8960.jpg
Processing file 275/284: water_body_897.jpg
Processing file 276/284: water_body_900.jpg
Processing file 277/284: water_body_938.jpg
Processing file 278/284: water_body_957.jpg
Processing file 279/284: water_body_967.jpg
Processing file 280/284: water_body_970.jpg
Processing file 281/284: water_body_978.jpg
Processing file 282/284: water_body_979.jpg
Processing file 283/284: water_body_987.jpg
Processing file 284/284: water_body_99.jpg
Batch processing completed. Results saved to waterbody-dataset\pred_val_frac_20
[eval] matched 284/284 files in waterbody-dataset\pred_val_frac_20
Found 284 image files to process
Processing file 1/284: water_body_1003.jpg
Processing file 2/284: water_body_1015.jpg
Processing file 3/284: water_body_105.jpg
Processing file 4/284: water_body_1052.jpg
Processing file 5/284: water_body_1059.jpg
Processing file 6/284: water_body_1062.jpg
Processing file 7/284: water_body_1074.jpg
Processing file 8/284: water_body_1076.jpg
Processing file 9/284: water_body_1089.jpg
Processing file 10/284: water_body_1096.jpg
Processing file 11/284: water_body_1104.jpg
Processing file 12/284: water_body_1109.jpg
Processing file 13/284: water_body_1112.jpg
Processing file 14/284: water_body_1124.jpg
Processing file 15/284: water_body_1184.jpg
Processing file 16/284: water_body_1203.jpg
Processing file 17/284: water_body_121.jpg
Processing file 18/284: water_body_1227.jpg
Processing file 19/284: water_body_1280.jpg
Processing file 20/284: water_body_1307.jpg
Processing file 21/284: water_body_1354.jpg
Processing file 22/284: water_body_1359.jpg
Processing file 23/284: water_body_1361.jpg
Processing file 24/284: water_body_1368.jpg
Processing file 25/284: water_body_1373.jpg
Processing file 26/284: water_body_1382.jpg
Processing file 27/284: water_body_1392.jpg
Processing file 28/284: water_body_1406.jpg
Processing file 29/284: water_body_1420.jpg
Processing file 30/284: water_body_1433.jpg
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Processing file 31/284: water_body_1439.jpg
Processing file 32/284: water_body_1451.jpg
Processing file 33/284: water_body_1468.jpg
Processing file 34/284: water_body_1502.jpg
Processing file 35/284: water_body_1513.jpg
Processing file 36/284: water_body_1525.jpg
Processing file 37/284: water_body_1559.jpg
Processing file 38/284: water_body_1565.jpg
Processing file 39/284: water_body_1572.jpg
Processing file 40/284: water_body_1577.jpg
Processing file 41/284: water_body_1579.jpg
Processing file 42/284: water_body_1586.jpg
Processing file 43/284: water_body_1591.jpg
Processing file 44/284: water_body_1593.jpg
Processing file 45/284: water_body_1622.jpg
Processing file 46/284: water_body_1623.jpg
Processing file 47/284: water_body_1626.jpg
Processing file 48/284: water_body_1630.jpg
Processing file 49/284: water_body_1655.jpg
Processing file 50/284: water_body_1658.jpg
Processing file 51/284: water_body_1662.jpg
Processing file 52/284: water_body_1663.jpg
Processing file 53/284: water_body_1664.jpg
Processing file 54/284: water_body_1684.jpg
Processing file 55/284: water_body_1693.jpg
Processing file 56/284: water_body_1733.jpg
Processing file 57/284: water_body_1760.jpg
Processing file 58/284: water_body_1768.jpg
Processing file 59/284: water_body_1769.jpg
Processing file 60/284: water_body_1787.jpg
Processing file 61/284: water_body_1796.jpg
Processing file 62/284: water_body_1801.jpg
Processing file 63/284: water_body_1813.jpg
Processing file 64/284: water_body_1815.jpg
Processing file 65/284: water_body_1831.jpg
Processing file 66/284: water_body_1844.jpg
Processing file 67/284: water_body_1847.jpg
Processing file 68/284: water_body_21.jpg
Processing file 69/284: water_body_2171.jpg
Processing file 70/284: water_body_2185.jpg
Processing file 71/284: water_body_2190.jpg
Processing file 72/284: water_body_22.jpg
Processing file 73/284: water_body_2217.jpg
Processing file 74/284: water_body_2255.jpg
Processing file 75/284: water_body_2264.jpg
Processing file 76/284: water_body_2270.jpg
Processing file 77/284: water_body_2282.jpg
Processing file 78/284: water_body_2285.jpg
Processing file 79/284: water_body_2293.jpg
Processing file 80/284: water_body_2295.jpg
Processing file 81/284: water_body_2299.jpg
Processing file 82/284: water_body_2307.jpg
Processing file 83/284: water_body_232.jpg
Processing file 84/284: water_body_2320.jpg
Processing file 85/284: water_body_2333.jpg
Processing file 86/284: water_body_2385.jpg
Processing file 87/284: water_body_2402.jpg
Processing file 88/284: water_body_2472.jpg
Processing file 89/284: water_body_2474.jpg
Processing file 90/284: water_body_2498.jpg
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Processing file 91/284: water_body_2520.jpg
Processing file 92/284: water_body_2529.jpg
Processing file 93/284: water_body_2538.jpg
Processing file 94/284: water_body_2541.jpg
Processing file 95/284: water_body_2553.jpg
Processing file 96/284: water_body_2559.jpg
Processing file 97/284: water_body_27.jpg
Processing file 98/284: water_body_39.jpg
Processing file 99/284: water_body_55.jpg
Processing file 100/284: water_body_590.jpg
Processing file 101/284: water_body_599.jpg
Processing file 102/284: water_body_607.jpg
Processing file 103/284: water_body_615.jpg
Processing file 104/284: water_body_620.jpg
Processing file 105/284: water_body_6377.jpg
Processing file 106/284: water_body_6385.jpg
Processing file 107/284: water_body_64.jpg
Processing file 108/284: water_body_6426.jpg
Processing file 109/284: water_body_668.jpg
Processing file 110/284: water_body_68.jpg
Processing file 111/284: water_body_690.jpg
Processing file 112/284: water_body_693.jpg
Processing file 113/284: water_body_7029.jpg
Processing file 114/284: water_body_704.jpg
Processing file 115/284: water_body_7040.jpg
Processing file 116/284: water_body_7052.jpg
Processing file 117/284: water_body_7057.jpg
Processing file 118/284: water_body_7067.jpg
Processing file 119/284: water_body_7068.jpg
Processing file 120/284: water_body_7079.jpg
Processing file 121/284: water_body_7100.jpg
Processing file 122/284: water_body_7132.jpg
Processing file 123/284: water_body_714.jpg
Processing file 124/284: water_body_7166.jpg
Processing file 125/284: water_body_7168.jpg
Processing file 126/284: water_body_7173.jpg
Processing file 127/284: water_body_7181.jpg
Processing file 128/284: water_body_7191.jpg
Processing file 129/284: water_body_7199.jpg
Processing file 130/284: water_body_72.jpg
Processing file 131/284: water_body_7207.jpg
Processing file 132/284: water_body_7235.jpg
Processing file 133/284: water_body_7238.jpg
Processing file 134/284: water_body_7245.jpg
Processing file 135/284: water_body_7247.jpg
Processing file 136/284: water_body_7252.jpg
Processing file 137/284: water_body_7265.jpg
Processing file 138/284: water_body_7275.jpg
Processing file 139/284: water_body_7276.jpg
Processing file 140/284: water_body_7278.jpg
Processing file 141/284: water_body_7289.jpg
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Processing file 144/284: water_body_7305.jpg
Processing file 145/284: water_body_7327.jpg
Processing file 146/284: water_body_7386.jpg
Processing file 147/284: water_body_7392.jpg
Processing file 148/284: water_body_7395.jpg
Processing file 149/284: water_body_7396.jpg
Processing file 150/284: water_body_7401.jpg
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Processing file 151/284: water_body_7413.jpg
Processing file 152/284: water_body_7416.jpg
Processing file 153/284: water_body_7417.jpg
Processing file 154/284: water_body_7418.jpg
Processing file 155/284: water_body_742.jpg
Processing file 156/284: water_body_7421.jpg
Processing file 157/284: water_body_7423.jpg
Processing file 158/284: water_body_7433.jpg
Processing file 159/284: water_body_7438.jpg
Processing file 160/284: water_body_744.jpg
Processing file 161/284: water_body_7486.jpg
Processing file 162/284: water_body_7499.jpg
Processing file 163/284: water_body_7510.jpg
Processing file 164/284: water_body_7511.jpg
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Processing file 168/284: water_body_7575.jpg
Processing file 169/284: water_body_7578.jpg
Processing file 170/284: water_body_7581.jpg
Processing file 171/284: water_body_7586.jpg
Processing file 172/284: water_body_759.jpg
Processing file 173/284: water_body_7603.jpg
Processing file 174/284: water_body_7615.jpg
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Processing file 178/284: water_body_7673.jpg
Processing file 179/284: water_body_7701.jpg
Processing file 180/284: water_body_771.jpg
Processing file 181/284: water_body_7712.jpg
Processing file 182/284: water_body_7717.jpg
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Processing file 185/284: water_body_78.jpg
Processing file 186/284: water_body_7810.jpg
Processing file 187/284: water_body_7843.jpg
Processing file 188/284: water_body_7871.jpg
Processing file 189/284: water_body_788.jpg
Processing file 190/284: water_body_7891.jpg
Processing file 191/284: water_body_790.jpg
Processing file 192/284: water_body_7901.jpg
Processing file 193/284: water_body_7914.jpg
Processing file 194/284: water_body_7918.jpg
Processing file 195/284: water_body_7956.jpg
Processing file 196/284: water_body_796.jpg
Processing file 197/284: water_body_7961.jpg
Processing file 198/284: water_body_7975.jpg
Processing file 199/284: water_body_7977.jpg
Processing file 200/284: water_body_800.jpg
Processing file 201/284: water_body_8005.jpg
Processing file 202/284: water_body_801.jpg
Processing file 203/284: water_body_8018.jpg
Processing file 204/284: water_body_8033.jpg
Processing file 205/284: water_body_8052.jpg
Processing file 206/284: water_body_8064.jpg
Processing file 207/284: water_body_8140.jpg
Processing file 208/284: water_body_8146.jpg
Processing file 209/284: water_body_8156.jpg
Processing file 210/284: water_body_8159.jpg
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Processing file 211/284: water_body_8169.jpg
Processing file 212/284: water_body_8195.jpg
Processing file 213/284: water_body_8206.jpg
Processing file 214/284: water_body_8245.jpg
Processing file 215/284: water_body_825.jpg
Processing file 216/284: water_body_8254.jpg
Processing file 217/284: water_body_8273.jpg
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Processing file 219/284: water_body_8321.jpg
Processing file 220/284: water_body_8352.jpg
Processing file 221/284: water_body_8367.jpg
Processing file 222/284: water_body_8376.jpg
Processing file 223/284: water_body_838.jpg
Processing file 224/284: water_body_8388.jpg
Processing file 225/284: water_body_8402.jpg
Processing file 226/284: water_body_8405.jpg
Processing file 227/284: water_body_841.jpg
Processing file 228/284: water_body_8446.jpg
Processing file 229/284: water_body_8448.jpg
Processing file 230/284: water_body_8463.jpg
Processing file 231/284: water_body_8467.jpg
Processing file 232/284: water_body_8482.jpg
Processing file 233/284: water_body_8491.jpg
Processing file 234/284: water_body_8495.jpg
Processing file 235/284: water_body_8501.jpg
Processing file 236/284: water_body_8504.jpg
Processing file 237/284: water_body_8506.jpg
Processing file 238/284: water_body_8509.jpg
Processing file 239/284: water_body_851.jpg
Processing file 240/284: water_body_8526.jpg
Processing file 241/284: water_body_8533.jpg
Processing file 242/284: water_body_8546.jpg
Processing file 243/284: water_body_8556.jpg
Processing file 244/284: water_body_8561.jpg
Processing file 245/284: water_body_8573.jpg
Processing file 246/284: water_body_8584.jpg
Processing file 247/284: water_body_8616.jpg
Processing file 248/284: water_body_8632.jpg
Processing file 249/284: water_body_8636.jpg
Processing file 250/284: water_body_8641.jpg
Processing file 251/284: water_body_8655.jpg
Processing file 252/284: water_body_8659.jpg
Processing file 253/284: water_body_8676.jpg
Processing file 254/284: water_body_8678.jpg
Processing file 255/284: water_body_8679.jpg
Processing file 256/284: water_body_8694.jpg
Processing file 257/284: water_body_8717.jpg
Processing file 258/284: water_body_8728.jpg
Processing file 259/284: water_body_8730.jpg
Processing file 260/284: water_body_8738.jpg
Processing file 261/284: water_body_8749.jpg
Processing file 262/284: water_body_8752.jpg
Processing file 263/284: water_body_8753.jpg
Processing file 264/284: water_body_8770.jpg
Processing file 265/284: water_body_8774.jpg
Processing file 266/284: water_body_8787.jpg
Processing file 267/284: water_body_8792.jpg
Processing file 268/284: water_body_881.jpg
Processing file 269/284: water_body_8847.jpg
Processing file 270/284: water_body_8850.jpg
```

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Processing file 271/284: water_body_8851.jpg
Processing file 272/284: water_body_8859.jpg
Processing file 273/284: water_body_889.jpg
Processing file 274/284: water_body_8912.jpg
Processing file 275/284: water_body_8921.jpg
Processing file 276/284: water_body_8926.jpg
Processing file 277/284: water_body_8934.jpg
Processing file 278/284: water_body_8935.jpg
Processing file 279/284: water_body_918.jpg
Processing file 280/284: water_body_949.jpg
Processing file 281/284: water_body_96.jpg
Processing file 282/284: water_body_964.jpg
Processing file 283/284: water_body_968.jpg
Processing file 284/284: water_body_994.jpg
Batch processing completed. Results saved to waterbody-dataset\pred_test_frac_20
[eval] matched 284/284 files in waterbody-dataset\pred_test_frac_20
VAL  -> OA:0.7170 | IoU:0.2855 | Dice:0.3545
TEST -> OA:0.7072 | IoU:0.2995 | Dice:0.3732
Model params: total=24,436,514 | trainable=24,436,514 | size=93.38 MB
Train time: 153.8s | Inference: 224.3 ms/img

=== Fraction 30% ===
Training subset size: 682
Internal val_split used: 0.0040
Using device: cuda
Found 682 image files and 682 label files
Training on 679 images, validating on 3 images
Testing data loader...
Data loader test passed.
Starting training with unet + resnet34
Model parameters: 24,436,514
Epoch: 1, Batch: 1/43, Loss: 0.6679, Time: 0.79s
Epoch: 1, Batch: 11/43, Loss: 0.4786, Time: 5.17s
Epoch: 1, Batch: 21/43, Loss: 0.4849, Time: 5.05s
Epoch: 1, Batch: 31/43, Loss: 0.4352, Time: 5.03s
Epoch: 1, Batch: 41/43, Loss: 0.4763, Time: 4.15s
Epoch 1/10: Train Loss: 0.5060, Val Loss: 5.5662, Val IoU: 0.2533, Val Dice: 0.31
34
Saving best model with IoU: 0.2533
Epoch: 2, Batch: 1/43, Loss: 0.4829, Time: 0.38s
Epoch: 2, Batch: 11/43, Loss: 0.5354, Time: 5.28s
Epoch: 2, Batch: 21/43, Loss: 0.3836, Time: 4.29s
Epoch: 2, Batch: 31/43, Loss: 0.4874, Time: 4.95s
Epoch: 2, Batch: 41/43, Loss: 0.4124, Time: 4.70s
Epoch 2/10: Train Loss: 0.4608, Val Loss: 0.2823, Val IoU: 0.6273, Val Dice: 0.70
53
Saving best model with IoU: 0.6273
Epoch: 3, Batch: 1/43, Loss: 0.5177, Time: 0.35s
Epoch: 3, Batch: 11/43, Loss: 0.4504, Time: 4.87s
Epoch: 3, Batch: 21/43, Loss: 0.4348, Time: 4.67s
Epoch: 3, Batch: 31/43, Loss: 0.4481, Time: 4.35s
Epoch: 3, Batch: 41/43, Loss: 0.3938, Time: 4.90s
Epoch 3/10: Train Loss: 0.4508, Val Loss: 0.3879, Val IoU: 0.5556, Val Dice: 0.65
05
Epoch: 4, Batch: 1/43, Loss: 0.5806, Time: 0.34s
Epoch: 4, Batch: 11/43, Loss: 0.3788, Time: 5.12s
Epoch: 4, Batch: 21/43, Loss: 0.4076, Time: 5.31s
Epoch: 4, Batch: 31/43, Loss: 0.3480, Time: 4.29s
Epoch: 4, Batch: 41/43, Loss: 0.4148, Time: 4.61s
Epoch 4/10: Train Loss: 0.4352, Val Loss: 0.3288, Val IoU: 0.6044, Val Dice: 0.69
```

```
04
Epoch: 5, Batch: 1/43, Loss: 0.3624, Time: 0.45s
Epoch: 5, Batch: 11/43, Loss: 0.4273, Time: 5.27s
Epoch: 5, Batch: 21/43, Loss: 0.3922, Time: 4.69s
Epoch: 5, Batch: 31/43, Loss: 0.3676, Time: 4.61s
Epoch: 5, Batch: 41/43, Loss: 0.3421, Time: 5.05s
Epoch 5/10: Train Loss: 0.4113, Val Loss: 0.3671, Val IoU: 0.5760, Val Dice: 0.66
53
Epoch: 6, Batch: 1/43, Loss: 0.3221, Time: 0.40s
Epoch: 6, Batch: 11/43, Loss: 0.5585, Time: 4.81s
Epoch: 6, Batch: 21/43, Loss: 0.4736, Time: 4.87s
Epoch: 6, Batch: 31/43, Loss: 0.4381, Time: 4.81s
Epoch: 6, Batch: 41/43, Loss: 0.3056, Time: 4.89s
Epoch 6/10: Train Loss: 0.4095, Val Loss: 0.3468, Val IoU: 0.5783, Val Dice: 0.67
16
Epoch: 7, Batch: 1/43, Loss: 0.4179, Time: 0.39s
Epoch: 7, Batch: 11/43, Loss: 0.4989, Time: 4.68s
Epoch: 7, Batch: 21/43, Loss: 0.4278, Time: 5.16s
Epoch: 7, Batch: 31/43, Loss: 0.3047, Time: 4.64s
Epoch: 7, Batch: 41/43, Loss: 0.3141, Time: 4.66s
Epoch 7/10: Train Loss: 0.3958, Val Loss: 0.3495, Val IoU: 0.6342, Val Dice: 0.71
65
Saving best model with IoU: 0.6342
Epoch: 8, Batch: 1/43, Loss: 0.3716, Time: 0.51s
Epoch: 8, Batch: 11/43, Loss: 0.4965, Time: 4.59s
Epoch: 8, Batch: 21/43, Loss: 0.4624, Time: 5.01s
Epoch: 8, Batch: 31/43, Loss: 0.3676, Time: 4.51s
Epoch: 8, Batch: 41/43, Loss: 0.3612, Time: 5.48s
Epoch 8/10: Train Loss: 0.3859, Val Loss: 0.3073, Val IoU: 0.6216, Val Dice: 0.70
84
Epoch: 9, Batch: 1/43, Loss: 0.2913, Time: 0.48s
Epoch: 9, Batch: 11/43, Loss: 0.3262, Time: 3.98s
Epoch: 9, Batch: 21/43, Loss: 0.2905, Time: 4.81s
Epoch: 9, Batch: 31/43, Loss: 0.3924, Time: 5.14s
Epoch: 9, Batch: 41/43, Loss: 0.4080, Time: 5.47s
Epoch 9/10: Train Loss: 0.3727, Val Loss: 0.2834, Val IoU: 0.6484, Val Dice: 0.72
57
Saving best model with IoU: 0.6484
Epoch: 10, Batch: 1/43, Loss: 0.4718, Time: 0.43s
Epoch: 10, Batch: 11/43, Loss: 0.3346, Time: 4.50s
Epoch: 10, Batch: 21/43, Loss: 0.3861, Time: 4.91s
Epoch: 10, Batch: 31/43, Loss: 0.4106, Time: 4.37s
Epoch: 10, Batch: 41/43, Loss: 0.2939, Time: 5.66s
Epoch 10/10: Train Loss: 0.3564, Val Loss: 0.2626, Val IoU: 0.6620, Val Dice: 0.7
335
Saving best model with IoU: 0.6620
Training complete! Best IoU: 0.6620
Models saved to waterbody-dataset\unet_resnet34_frac_30_models
Found 284 image files to process
Processing file 1/284: water_body_1044.jpg
Processing file 2/284: water_body_1064.jpg
Processing file 3/284: water_body_107.jpg
Processing file 4/284: water_body_1070.jpg
Processing file 5/284: water_body_11.jpg
Processing file 6/284: water_body_1133.jpg
Processing file 7/284: water_body_1143.jpg
Processing file 8/284: water_body_1152.jpg
Processing file 9/284: water_body_1170.jpg
Processing file 10/284: water_body_118.jpg
Processing file 11/284: water_body_1195.jpg
```

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Processing file 12/284: water_body_120.jpg
Processing file 13/284: water_body_1201.jpg
Processing file 14/284: water_body_1204.jpg
Processing file 15/284: water_body_1228.jpg
Processing file 16/284: water_body_1235.jpg
Processing file 17/284: water_body_1237.jpg
Processing file 18/284: water_body_1253.jpg
Processing file 19/284: water_body_1260.jpg
Processing file 20/284: water_body_1313.jpg
Processing file 21/284: water_body_1329.jpg
Processing file 22/284: water_body_1334.jpg
Processing file 23/284: water_body_1343.jpg
Processing file 24/284: water_body_1347.jpg
Processing file 25/284: water_body_1369.jpg
Processing file 26/284: water_body_1383.jpg
Processing file 27/284: water_body_1417.jpg
Processing file 28/284: water_body_1422.jpg
Processing file 29/284: water_body_1447.jpg
Processing file 30/284: water_body_1453.jpg
Processing file 31/284: water_body_1459.jpg
Processing file 32/284: water_body_1462.jpg
Processing file 33/284: water_body_1470.jpg
Processing file 34/284: water_body_1482.jpg
Processing file 35/284: water_body_1498.jpg
Processing file 36/284: water_body_1509.jpg
Processing file 37/284: water_body_1510.jpg
Processing file 38/284: water_body_1511.jpg
Processing file 39/284: water_body_1529.jpg
Processing file 40/284: water_body_1532.jpg
Processing file 41/284: water_body_1553.jpg
Processing file 42/284: water_body_1575.jpg
Processing file 43/284: water_body_1582.jpg
Processing file 44/284: water_body_1594.jpg
Processing file 45/284: water_body_1597.jpg
Processing file 46/284: water_body_1611.jpg
Processing file 47/284: water_body_1615.jpg
Processing file 48/284: water_body_1618.jpg
Processing file 49/284: water_body_1620.jpg
Processing file 50/284: water_body_1628.jpg
Processing file 51/284: water_body_1634.jpg
Processing file 52/284: water_body_1645.jpg
Processing file 53/284: water_body_1656.jpg
Processing file 54/284: water_body_1657.jpg
Processing file 55/284: water_body_1674.jpg
Processing file 56/284: water_body_1688.jpg
Processing file 57/284: water_body_1691.jpg
Processing file 58/284: water_body_1694.jpg
Processing file 59/284: water_body_1699.jpg
Processing file 60/284: water_body_1703.jpg
Processing file 61/284: water_body_1704.jpg
Processing file 62/284: water_body_1711.jpg
Processing file 63/284: water_body_1724.jpg
Processing file 64/284: water_body_1734.jpg
Processing file 65/284: water_body_1745.jpg
Processing file 66/284: water_body_1749.jpg
Processing file 67/284: water_body_1770.jpg
Processing file 68/284: water_body_1771.jpg
Processing file 69/284: water_body_1776.jpg
Processing file 70/284: water_body_1794.jpg
Processing file 71/284: water_body_1816.jpg
```

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Processing file 72/284: water_body_1832.jpg
Processing file 73/284: water_body_1833.jpg
Processing file 74/284: water_body_1839.jpg
Processing file 75/284: water_body_186.jpg
Processing file 76/284: water_body_2182.jpg
Processing file 77/284: water_body_2203.jpg
Processing file 78/284: water_body_2220.jpg
Processing file 79/284: water_body_2227.jpg
Processing file 80/284: water_body_2235.jpg
Processing file 81/284: water_body_2246.jpg
Processing file 82/284: water_body_2250.jpg
Processing file 83/284: water_body_2253.jpg
Processing file 84/284: water_body_2276.jpg
Processing file 85/284: water_body_2287.jpg
Processing file 86/284: water_body_2291.jpg
Processing file 87/284: water_body_2340.jpg
Processing file 88/284: water_body_2370.jpg
Processing file 89/284: water_body_2372.jpg
Processing file 90/284: water_body_2373.jpg
Processing file 91/284: water_body_2395.jpg
Processing file 92/284: water_body_2397.jpg
Processing file 93/284: water_body_2415.jpg
Processing file 94/284: water_body_2432.jpg
Processing file 95/284: water_body_2440.jpg
Processing file 96/284: water_body_2456.jpg
Processing file 97/284: water_body_2500.jpg
Processing file 98/284: water_body_2504.jpg
Processing file 99/284: water_body_2527.jpg
Processing file 100/284: water_body_2530.jpg
Processing file 101/284: water_body_2532.jpg
Processing file 102/284: water_body_2533.jpg
Processing file 103/284: water_body_2554.jpg
Processing file 104/284: water_body_2561.jpg
Processing file 105/284: water_body_29.jpg
Processing file 106/284: water_body_42.jpg
Processing file 107/284: water_body_59.jpg
Processing file 108/284: water_body_606.jpg
Processing file 109/284: water_body_631.jpg
Processing file 110/284: water_body_6378.jpg
Processing file 111/284: water_body_6389.jpg
Processing file 112/284: water_body_6421.jpg
Processing file 113/284: water_body_644.jpg
Processing file 114/284: water_body_65.jpg
Processing file 115/284: water_body_654.jpg
Processing file 116/284: water_body_7037.jpg
Processing file 117/284: water_body_7045.jpg
Processing file 118/284: water_body_7056.jpg
Processing file 119/284: water_body_7058.jpg
Processing file 120/284: water_body_7071.jpg
Processing file 121/284: water_body_7090.jpg
Processing file 122/284: water_body_7095.jpg
Processing file 123/284: water_body_71.jpg
Processing file 124/284: water_body_7137.jpg
Processing file 125/284: water_body_7140.jpg
Processing file 126/284: water_body_7147.jpg
Processing file 127/284: water_body_7162.jpg
Processing file 128/284: water_body_7165.jpg
Processing file 129/284: water_body_7172.jpg
Processing file 130/284: water_body_718.jpg
Processing file 131/284: water_body_7184.jpg
```

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Processing file 132/284: water_body_7192.jpg
Processing file 133/284: water_body_720.jpg
Processing file 134/284: water_body_7205.jpg
Processing file 135/284: water_body_7228.jpg
Processing file 136/284: water_body_7230.jpg
Processing file 137/284: water_body_7232.jpg
Processing file 138/284: water_body_7244.jpg
Processing file 139/284: water_body_7258.jpg
Processing file 140/284: water_body_7260.jpg
Processing file 141/284: water_body_7262.jpg
Processing file 142/284: water_body_7283.jpg
Processing file 143/284: water_body_7319.jpg
Processing file 144/284: water_body_7333.jpg
Processing file 145/284: water_body_7352.jpg
Processing file 146/284: water_body_7404.jpg
Processing file 147/284: water_body_7420.jpg
Processing file 148/284: water_body_7429.jpg
Processing file 149/284: water_body_7432.jpg
Processing file 150/284: water_body_7436.jpg
Processing file 151/284: water_body_7437.jpg
Processing file 152/284: water_body_7444.jpg
Processing file 153/284: water_body_7475.jpg
Processing file 154/284: water_body_7508.jpg
Processing file 155/284: water_body_7515.jpg
Processing file 156/284: water_body_7517.jpg
Processing file 157/284: water_body_7518.jpg
Processing file 158/284: water_body_7525.jpg
Processing file 159/284: water_body_7540.jpg
Processing file 160/284: water_body_7544.jpg
Processing file 161/284: water_body_7569.jpg
Processing file 162/284: water_body_7626.jpg
Processing file 163/284: water_body_7686.jpg
Processing file 164/284: water_body_7693.jpg
Processing file 165/284: water_body_7696.jpg
Processing file 166/284: water_body_7703.jpg
Processing file 167/284: water_body_7707.jpg
Processing file 168/284: water_body_7708.jpg
Processing file 169/284: water_body_7709.jpg
Processing file 170/284: water_body_7711.jpg
Processing file 171/284: water_body_772.jpg
Processing file 172/284: water_body_7720.jpg
Processing file 173/284: water_body_7727.jpg
Processing file 174/284: water_body_7729.jpg
Processing file 175/284: water_body_773.jpg
Processing file 176/284: water_body_7750.jpg
Processing file 177/284: water_body_7773.jpg
Processing file 178/284: water_body_7781.jpg
Processing file 179/284: water_body_7791.jpg
Processing file 180/284: water_body_7815.jpg
Processing file 181/284: water_body_7823.jpg
Processing file 182/284: water_body_7828.jpg
Processing file 183/284: water_body_783.jpg
Processing file 184/284: water_body_7848.jpg
Processing file 185/284: water_body_7868.jpg
Processing file 186/284: water_body_7869.jpg
Processing file 187/284: water_body_7892.jpg
Processing file 188/284: water_body_7903.jpg
Processing file 189/284: water_body_7944.jpg
Processing file 190/284: water_body_7946.jpg
Processing file 191/284: water_body_7947.jpg
```

```
Processing file 192/284: water_body_7954.jpg
Processing file 193/284: water_body_7955.jpg
Processing file 194/284: water_body_7957.jpg
Processing file 195/284: water_body_7978.jpg
Processing file 196/284: water_body_8002.jpg
Processing file 197/284: water_body_8013.jpg
Processing file 198/284: water_body_8017.jpg
Processing file 199/284: water_body_8027.jpg
Processing file 200/284: water_body_8059.jpg
Processing file 201/284: water_body_806.jpg
Processing file 202/284: water_body_8068.jpg
Processing file 203/284: water_body_8076.jpg
Processing file 204/284: water_body_8098.jpg
Processing file 205/284: water_body_810.jpg
Processing file 206/284: water_body_811.jpg
Processing file 207/284: water_body_8165.jpg
Processing file 208/284: water_body_8177.jpg
Processing file 209/284: water_body_818.jpg
Processing file 210/284: water_body_8185.jpg
Processing file 211/284: water_body_8219.jpg
Processing file 212/284: water_body_823.jpg
Processing file 213/284: water_body_8242.jpg
Processing file 214/284: water_body_8248.jpg
Processing file 215/284: water_body_8253.jpg
Processing file 216/284: water_body_8319.jpg
Processing file 217/284: water_body_8338.jpg
Processing file 218/284: water_body_8344.jpg
Processing file 219/284: water_body_8348.jpg
Processing file 220/284: water_body_8356.jpg
Processing file 221/284: water_body_8377.jpg
Processing file 222/284: water_body_8387.jpg
Processing file 223/284: water_body_839.jpg
Processing file 224/284: water_body_8398.jpg
Processing file 225/284: water_body_8408.jpg
Processing file 226/284: water_body_8412.jpg
Processing file 227/284: water_body_8423.jpg
Processing file 228/284: water_body_8424.jpg
Processing file 229/284: water_body_8439.jpg
Processing file 230/284: water_body_8442.jpg
Processing file 231/284: water_body_8455.jpg
Processing file 232/284: water_body_8462.jpg
Processing file 233/284: water_body_847.jpg
Processing file 234/284: water_body_8476.jpg
Processing file 235/284: water_body_8497.jpg
Processing file 236/284: water_body_8558.jpg
Processing file 237/284: water_body_8560.jpg
Processing file 238/284: water_body_8583.jpg
Processing file 239/284: water_body_8607.jpg
Processing file 240/284: water_body_8619.jpg
Processing file 241/284: water_body_8622.jpg
Processing file 242/284: water_body_8633.jpg
Processing file 243/284: water_body_8650.jpg
Processing file 244/284: water_body_8651.jpg
Processing file 245/284: water_body_8653.jpg
Processing file 246/284: water_body_8664.jpg
Processing file 247/284: water_body_8670.jpg
Processing file 248/284: water_body_8671.jpg
Processing file 249/284: water_body_8680.jpg
Processing file 250/284: water_body_8687.jpg
Processing file 251/284: water_body_8693.jpg
```

```
Processing file 252/284: water_body_8702.jpg
Processing file 253/284: water_body_8703.jpg
Processing file 254/284: water_body_8715.jpg
Processing file 255/284: water_body_8722.jpg
Processing file 256/284: water_body_8729.jpg
Processing file 257/284: water_body_875.jpg
Processing file 258/284: water_body_8760.jpg
Processing file 259/284: water_body_8763.jpg
Processing file 260/284: water_body_8764.jpg
Processing file 261/284: water_body_8789.jpg
Processing file 262/284: water_body_8823.jpg
Processing file 263/284: water_body_8825.jpg
Processing file 264/284: water_body_8843.jpg
Processing file 265/284: water_body_885.jpg
Processing file 266/284: water_body_8854.jpg
Processing file 267/284: water_body_8877.jpg
Processing file 268/284: water_body_8885.jpg
Processing file 269/284: water_body_8911.jpg
Processing file 270/284: water_body_8918.jpg
Processing file 271/284: water_body_8930.jpg
Processing file 272/284: water_body_8949.jpg
Processing file 273/284: water_body_8953.jpg
Processing file 274/284: water_body_8960.jpg
Processing file 275/284: water_body_897.jpg
Processing file 276/284: water_body_900.jpg
Processing file 277/284: water_body_938.jpg
Processing file 278/284: water_body_957.jpg
Processing file 279/284: water_body_967.jpg
Processing file 280/284: water_body_970.jpg
Processing file 281/284: water_body_978.jpg
Processing file 282/284: water_body_979.jpg
Processing file 283/284: water_body_987.jpg
Processing file 284/284: water_body_99.jpg
Batch processing completed. Results saved to waterbody-dataset\pred_val_frac_30
[eval] matched 284/284 files in waterbody-dataset\pred_val_frac_30
Found 284 image files to process
Processing file 1/284: water_body_1003.jpg
Processing file 2/284: water_body_1015.jpg
Processing file 3/284: water_body_105.jpg
Processing file 4/284: water_body_1052.jpg
Processing file 5/284: water_body_1059.jpg
Processing file 6/284: water_body_1062.jpg
Processing file 7/284: water_body_1074.jpg
Processing file 8/284: water_body_1076.jpg
Processing file 9/284: water_body_1089.jpg
Processing file 10/284: water_body_1096.jpg
Processing file 11/284: water_body_1104.jpg
Processing file 12/284: water_body_1109.jpg
Processing file 13/284: water_body_1112.jpg
Processing file 14/284: water_body_1124.jpg
Processing file 15/284: water_body_1184.jpg
Processing file 16/284: water_body_1203.jpg
Processing file 17/284: water_body_121.jpg
Processing file 18/284: water_body_1227.jpg
Processing file 19/284: water_body_1280.jpg
Processing file 20/284: water_body_1307.jpg
Processing file 21/284: water_body_1354.jpg
Processing file 22/284: water_body_1359.jpg
Processing file 23/284: water_body_1361.jpg
Processing file 24/284: water_body_1368.jpg
```

```
Processing file 25/284: water_body_1373.jpg
Processing file 26/284: water_body_1382.jpg
Processing file 27/284: water_body_1392.jpg
Processing file 28/284: water_body_1406.jpg
Processing file 29/284: water_body_1420.jpg
Processing file 30/284: water_body_1433.jpg
Processing file 31/284: water_body_1439.jpg
Processing file 32/284: water_body_1451.jpg
Processing file 33/284: water_body_1468.jpg
Processing file 34/284: water_body_1502.jpg
Processing file 35/284: water_body_1513.jpg
Processing file 36/284: water_body_1525.jpg
Processing file 37/284: water_body_1559.jpg
Processing file 38/284: water_body_1565.jpg
Processing file 39/284: water_body_1572.jpg
Processing file 40/284: water_body_1577.jpg
Processing file 41/284: water_body_1579.jpg
Processing file 42/284: water_body_1586.jpg
Processing file 43/284: water_body_1591.jpg
Processing file 44/284: water_body_1593.jpg
Processing file 45/284: water_body_1622.jpg
Processing file 46/284: water_body_1623.jpg
Processing file 47/284: water_body_1626.jpg
Processing file 48/284: water_body_1630.jpg
Processing file 49/284: water_body_1655.jpg
Processing file 50/284: water_body_1658.jpg
Processing file 51/284: water_body_1662.jpg
Processing file 52/284: water_body_1663.jpg
Processing file 53/284: water_body_1664.jpg
Processing file 54/284: water_body_1684.jpg
Processing file 55/284: water_body_1693.jpg
Processing file 56/284: water_body_1733.jpg
Processing file 57/284: water_body_1760.jpg
Processing file 58/284: water_body_1768.jpg
Processing file 59/284: water_body_1769.jpg
Processing file 60/284: water_body_1787.jpg
Processing file 61/284: water_body_1796.jpg
Processing file 62/284: water_body_1801.jpg
Processing file 63/284: water_body_1813.jpg
Processing file 64/284: water_body_1815.jpg
Processing file 65/284: water_body_1831.jpg
Processing file 66/284: water_body_1844.jpg
Processing file 67/284: water_body_1847.jpg
Processing file 68/284: water_body_21.jpg
Processing file 69/284: water_body_2171.jpg
Processing file 70/284: water_body_2185.jpg
Processing file 71/284: water_body_2190.jpg
Processing file 72/284: water_body_22.jpg
Processing file 73/284: water_body_2217.jpg
Processing file 74/284: water_body_2255.jpg
Processing file 75/284: water_body_2264.jpg
Processing file 76/284: water_body_2270.jpg
Processing file 77/284: water_body_2282.jpg
Processing file 78/284: water_body_2285.jpg
Processing file 79/284: water_body_2293.jpg
Processing file 80/284: water_body_2295.jpg
Processing file 81/284: water_body_2299.jpg
Processing file 82/284: water_body_2307.jpg
Processing file 83/284: water_body_232.jpg
Processing file 84/284: water_body_2320.jpg
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Processing file 85/284: water_body_2333.jpg
Processing file 86/284: water_body_2385.jpg
Processing file 87/284: water_body_2402.jpg
Processing file 88/284: water_body_2472.jpg
Processing file 89/284: water_body_2474.jpg
Processing file 90/284: water_body_2498.jpg
Processing file 91/284: water_body_2520.jpg
Processing file 92/284: water_body_2529.jpg
Processing file 93/284: water_body_2538.jpg
Processing file 94/284: water_body_2541.jpg
Processing file 95/284: water_body_2553.jpg
Processing file 96/284: water_body_2559.jpg
Processing file 97/284: water_body_27.jpg
Processing file 98/284: water_body_39.jpg
Processing file 99/284: water_body_55.jpg
Processing file 100/284: water_body_590.jpg
Processing file 101/284: water_body_599.jpg
Processing file 102/284: water_body_607.jpg
Processing file 103/284: water_body_615.jpg
Processing file 104/284: water_body_620.jpg
Processing file 105/284: water_body_6377.jpg
Processing file 106/284: water_body_6385.jpg
Processing file 107/284: water_body_64.jpg
Processing file 108/284: water_body_6426.jpg
Processing file 109/284: water_body_668.jpg
Processing file 110/284: water_body_68.jpg
Processing file 111/284: water_body_690.jpg
Processing file 112/284: water_body_693.jpg
Processing file 113/284: water_body_7029.jpg
Processing file 114/284: water_body_704.jpg
Processing file 115/284: water_body_7040.jpg
Processing file 116/284: water_body_7052.jpg
Processing file 117/284: water_body_7057.jpg
Processing file 118/284: water_body_7067.jpg
Processing file 119/284: water_body_7068.jpg
Processing file 120/284: water_body_7079.jpg
Processing file 121/284: water_body_7100.jpg
Processing file 122/284: water_body_7132.jpg
Processing file 123/284: water_body_714.jpg
Processing file 124/284: water_body_7166.jpg
Processing file 125/284: water_body_7168.jpg
Processing file 126/284: water_body_7173.jpg
Processing file 127/284: water_body_7181.jpg
Processing file 128/284: water_body_7191.jpg
Processing file 129/284: water_body_7199.jpg
Processing file 130/284: water_body_72.jpg
Processing file 131/284: water_body_7207.jpg
Processing file 132/284: water_body_7235.jpg
Processing file 133/284: water_body_7238.jpg
Processing file 134/284: water_body_7245.jpg
Processing file 135/284: water_body_7247.jpg
Processing file 136/284: water_body_7252.jpg
Processing file 137/284: water_body_7265.jpg
Processing file 138/284: water_body_7275.jpg
Processing file 139/284: water_body_7276.jpg
Processing file 140/284: water_body_7278.jpg
Processing file 141/284: water_body_7289.jpg
Processing file 142/284: water_body_7290.jpg
Processing file 143/284: water_body_7296.jpg
Processing file 144/284: water_body_7305.jpg
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Processing file 145/284: water_body_7327.jpg
Processing file 146/284: water_body_7386.jpg
Processing file 147/284: water_body_7392.jpg
Processing file 148/284: water_body_7395.jpg
Processing file 149/284: water_body_7396.jpg
Processing file 150/284: water_body_7401.jpg
Processing file 151/284: water_body_7413.jpg
Processing file 152/284: water_body_7416.jpg
Processing file 153/284: water_body_7417.jpg
Processing file 154/284: water_body_7418.jpg
Processing file 155/284: water_body_742.jpg
Processing file 156/284: water_body_7421.jpg
Processing file 157/284: water_body_7423.jpg
Processing file 158/284: water_body_7433.jpg
Processing file 159/284: water_body_7438.jpg
Processing file 160/284: water_body_744.jpg
Processing file 161/284: water_body_7486.jpg
Processing file 162/284: water_body_7499.jpg
Processing file 163/284: water_body_7510.jpg
Processing file 164/284: water_body_7511.jpg
Processing file 165/284: water_body_7543.jpg
Processing file 166/284: water_body_7552.jpg
Processing file 167/284: water_body_7562.jpg
Processing file 168/284: water_body_7575.jpg
Processing file 169/284: water_body_7578.jpg
Processing file 170/284: water_body_7581.jpg
Processing file 171/284: water_body_7586.jpg
Processing file 172/284: water_body_759.jpg
Processing file 173/284: water_body_7603.jpg
Processing file 174/284: water_body_7615.jpg
Processing file 175/284: water_body_7618.jpg
Processing file 176/284: water_body_7629.jpg
Processing file 177/284: water_body_7638.jpg
Processing file 178/284: water_body_7673.jpg
Processing file 179/284: water_body_7701.jpg
Processing file 180/284: water_body_771.jpg
Processing file 181/284: water_body_7712.jpg
Processing file 182/284: water_body_7717.jpg
Processing file 183/284: water_body_7766.jpg
Processing file 184/284: water_body_7793.jpg
Processing file 185/284: water_body_78.jpg
Processing file 186/284: water_body_7810.jpg
Processing file 187/284: water_body_7843.jpg
Processing file 188/284: water_body_7871.jpg
Processing file 189/284: water_body_788.jpg
Processing file 190/284: water_body_7891.jpg
Processing file 191/284: water_body_790.jpg
Processing file 192/284: water_body_7901.jpg
Processing file 193/284: water_body_7914.jpg
Processing file 194/284: water_body_7918.jpg
Processing file 195/284: water_body_7956.jpg
Processing file 196/284: water_body_796.jpg
Processing file 197/284: water_body_7961.jpg
Processing file 198/284: water_body_7975.jpg
Processing file 199/284: water_body_7977.jpg
Processing file 200/284: water_body_800.jpg
Processing file 201/284: water_body_8005.jpg
Processing file 202/284: water_body_801.jpg
Processing file 203/284: water_body_8018.jpg
Processing file 204/284: water_body_8033.jpg
```

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Processing file 205/284: water_body_8052.jpg
Processing file 206/284: water_body_8064.jpg
Processing file 207/284: water_body_8140.jpg
Processing file 208/284: water_body_8146.jpg
Processing file 209/284: water_body_8156.jpg
Processing file 210/284: water_body_8159.jpg
Processing file 211/284: water_body_8169.jpg
Processing file 212/284: water_body_8195.jpg
Processing file 213/284: water_body_8206.jpg
Processing file 214/284: water_body_8245.jpg
Processing file 215/284: water_body_825.jpg
Processing file 216/284: water_body_8254.jpg
Processing file 217/284: water_body_8273.jpg
Processing file 218/284: water_body_8288.jpg
Processing file 219/284: water_body_8321.jpg
Processing file 220/284: water_body_8352.jpg
Processing file 221/284: water_body_8367.jpg
Processing file 222/284: water_body_8376.jpg
Processing file 223/284: water_body_838.jpg
Processing file 224/284: water_body_8388.jpg
Processing file 225/284: water_body_8402.jpg
Processing file 226/284: water_body_8405.jpg
Processing file 227/284: water_body_841.jpg
Processing file 228/284: water_body_8446.jpg
Processing file 229/284: water_body_8448.jpg
Processing file 230/284: water_body_8463.jpg
Processing file 231/284: water_body_8467.jpg
Processing file 232/284: water_body_8482.jpg
Processing file 233/284: water_body_8491.jpg
Processing file 234/284: water_body_8495.jpg
Processing file 235/284: water_body_8501.jpg
Processing file 236/284: water_body_8504.jpg
Processing file 237/284: water_body_8506.jpg
Processing file 238/284: water_body_8509.jpg
Processing file 239/284: water_body_851.jpg
Processing file 240/284: water_body_8526.jpg
Processing file 241/284: water_body_8533.jpg
Processing file 242/284: water_body_8546.jpg
Processing file 243/284: water_body_8556.jpg
Processing file 244/284: water_body_8561.jpg
Processing file 245/284: water_body_8573.jpg
Processing file 246/284: water_body_8584.jpg
Processing file 247/284: water_body_8616.jpg
Processing file 248/284: water_body_8632.jpg
Processing file 249/284: water_body_8636.jpg
Processing file 250/284: water_body_8641.jpg
Processing file 251/284: water_body_8655.jpg
Processing file 252/284: water_body_8659.jpg
Processing file 253/284: water_body_8676.jpg
Processing file 254/284: water_body_8678.jpg
Processing file 255/284: water_body_8679.jpg
Processing file 256/284: water_body_8694.jpg
Processing file 257/284: water_body_8717.jpg
Processing file 258/284: water_body_8728.jpg
Processing file 259/284: water_body_8730.jpg
Processing file 260/284: water_body_8738.jpg
Processing file 261/284: water_body_8749.jpg
Processing file 262/284: water_body_8752.jpg
Processing file 263/284: water_body_8753.jpg
Processing file 264/284: water_body_8770.jpg
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Processing file 265/284: water_body_8774.jpg
Processing file 266/284: water_body_8787.jpg
Processing file 267/284: water_body_8792.jpg
Processing file 268/284: water_body_881.jpg
Processing file 269/284: water_body_8847.jpg
Processing file 270/284: water_body_8850.jpg
Processing file 271/284: water_body_8851.jpg
Processing file 272/284: water_body_8859.jpg
Processing file 273/284: water_body_889.jpg
Processing file 274/284: water_body_8912.jpg
Processing file 275/284: water_body_8921.jpg
Processing file 276/284: water_body_8926.jpg
Processing file 277/284: water_body_8934.jpg
Processing file 278/284: water_body_8935.jpg
Processing file 279/284: water_body_918.jpg
Processing file 280/284: water_body_949.jpg
Processing file 281/284: water_body_96.jpg
Processing file 282/284: water_body_964.jpg
Processing file 283/284: water_body_968.jpg
Processing file 284/284: water_body_994.jpg
Batch processing completed. Results saved to waterbody-dataset\pred_test_frac_30
[eval] matched 284/284 files in waterbody-dataset\pred_test_frac_30
VAL  -> OA:0.8096 | IoU:0.3565 | Dice:0.4139
TEST -> OA:0.8036 | IoU:0.3843 | Dice:0.4456
Model params: total=24,436,514 | trainable=24,436,514 | size=93.38 MB
Train time: 207.0s | Inference: 228.6 ms/img

=== Fraction 50% ===
Training subset size: 1136
Internal val_split used: 0.0040
Using device: cuda
Found 1136 image files and 1136 label files
Training on 1131 images, validating on 5 images
Testing data loader...
Data loader test passed.
Starting training with unet + resnet34
Model parameters: 24,436,514
Epoch: 1, Batch: 1/71, Loss: 0.8026, Time: 0.72s
Epoch: 1, Batch: 11/71, Loss: 0.5734, Time: 5.14s
Epoch: 1, Batch: 21/71, Loss: 0.3880, Time: 5.71s
Epoch: 1, Batch: 31/71, Loss: 0.5477, Time: 6.16s
Epoch: 1, Batch: 41/71, Loss: 0.4452, Time: 6.00s
Epoch: 1, Batch: 51/71, Loss: 0.3701, Time: 4.66s
Epoch: 1, Batch: 61/71, Loss: 0.5975, Time: 4.73s
Epoch: 1, Batch: 71/71, Loss: 0.5840, Time: 4.81s
Epoch 1/10: Train Loss: 0.4917, Val Loss: 0.5341, Val IoU: 0.5856, Val Dice: 0.65
71
Saving best model with IoU: 0.5856
Epoch: 2, Batch: 1/71, Loss: 0.3934, Time: 0.52s
Epoch: 2, Batch: 11/71, Loss: 0.4238, Time: 5.61s
Epoch: 2, Batch: 21/71, Loss: 0.4502, Time: 5.42s
Epoch: 2, Batch: 31/71, Loss: 0.3519, Time: 4.85s
Epoch: 2, Batch: 41/71, Loss: 0.3468, Time: 5.26s
Epoch: 2, Batch: 51/71, Loss: 0.3986, Time: 6.47s
Epoch: 2, Batch: 61/71, Loss: 0.4331, Time: 5.23s
Epoch: 2, Batch: 71/71, Loss: 0.3904, Time: 4.36s
Epoch 2/10: Train Loss: 0.4574, Val Loss: 0.3750, Val IoU: 0.5597, Val Dice: 0.63
40
Epoch: 3, Batch: 1/71, Loss: 0.4296, Time: 0.39s
Epoch: 3, Batch: 11/71, Loss: 0.6053, Time: 6.13s
```

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Epoch: 3, Batch: 21/71, Loss: 0.5783, Time: 5.49s
Epoch: 3, Batch: 31/71, Loss: 0.3675, Time: 5.20s
Epoch: 3, Batch: 41/71, Loss: 0.3826, Time: 5.99s
Epoch: 3, Batch: 51/71, Loss: 0.3936, Time: 5.36s
Epoch: 3, Batch: 61/71, Loss: 0.3907, Time: 4.88s
Epoch: 3, Batch: 71/71, Loss: 0.3200, Time: 4.18s
Epoch 3/10: Train Loss: 0.4219, Val Loss: 0.7761, Val IoU: 0.5519, Val Dice: 0.65
45
Epoch: 4, Batch: 1/71, Loss: 0.3700, Time: 0.42s
Epoch: 4, Batch: 11/71, Loss: 0.3373, Time: 4.57s
Epoch: 4, Batch: 21/71, Loss: 0.4015, Time: 5.81s
Epoch: 4, Batch: 31/71, Loss: 0.4165, Time: 4.75s
Epoch: 4, Batch: 41/71, Loss: 0.4800, Time: 5.90s
Epoch: 4, Batch: 51/71, Loss: 0.3487, Time: 5.40s
Epoch: 4, Batch: 61/71, Loss: 0.4481, Time: 5.34s
Epoch: 4, Batch: 71/71, Loss: 0.5136, Time: 5.64s
Epoch 4/10: Train Loss: 0.4167, Val Loss: 0.3277, Val IoU: 0.6755, Val Dice: 0.75
94
Saving best model with IoU: 0.6755
Epoch: 5, Batch: 1/71, Loss: 0.3895, Time: 0.49s
Epoch: 5, Batch: 11/71, Loss: 0.3320, Time: 5.65s
Epoch: 5, Batch: 21/71, Loss: 0.3061, Time: 6.14s
Epoch: 5, Batch: 31/71, Loss: 0.3969, Time: 4.25s
Epoch: 5, Batch: 41/71, Loss: 0.5605, Time: 5.45s
Epoch: 5, Batch: 51/71, Loss: 0.3979, Time: 6.03s
Epoch: 5, Batch: 61/71, Loss: 0.3887, Time: 4.68s
Epoch: 5, Batch: 71/71, Loss: 0.3559, Time: 4.93s
Epoch 5/10: Train Loss: 0.3934, Val Loss: 0.4346, Val IoU: 0.5743, Val Dice: 0.67
59
Epoch: 6, Batch: 1/71, Loss: 0.3264, Time: 0.44s
Epoch: 6, Batch: 11/71, Loss: 0.2710, Time: 5.53s
Epoch: 6, Batch: 21/71, Loss: 0.3862, Time: 5.22s
Epoch: 6, Batch: 31/71, Loss: 0.4713, Time: 4.91s
Epoch: 6, Batch: 41/71, Loss: 0.3244, Time: 5.73s
Epoch: 6, Batch: 51/71, Loss: 0.3818, Time: 6.59s
Epoch: 6, Batch: 61/71, Loss: 0.3785, Time: 4.88s
Epoch: 6, Batch: 71/71, Loss: 0.4160, Time: 4.32s
Epoch 6/10: Train Loss: 0.3925, Val Loss: 0.2944, Val IoU: 0.6847, Val Dice: 0.76
54
Saving best model with IoU: 0.6847
Epoch: 7, Batch: 1/71, Loss: 0.4052, Time: 0.72s
Epoch: 7, Batch: 11/71, Loss: 0.3365, Time: 4.62s
Epoch: 7, Batch: 21/71, Loss: 0.3123, Time: 5.22s
Epoch: 7, Batch: 31/71, Loss: 0.5041, Time: 5.55s
Epoch: 7, Batch: 41/71, Loss: 0.5188, Time: 5.07s
Epoch: 7, Batch: 51/71, Loss: 0.4414, Time: 5.94s
Epoch: 7, Batch: 61/71, Loss: 0.4484, Time: 4.48s
Epoch: 7, Batch: 71/71, Loss: 0.4166, Time: 6.02s
Epoch 7/10: Train Loss: 0.3921, Val Loss: 0.3279, Val IoU: 0.6535, Val Dice: 0.74
36
Epoch: 8, Batch: 1/71, Loss: 0.3614, Time: 0.58s
Epoch: 8, Batch: 11/71, Loss: 0.4661, Time: 5.72s
Epoch: 8, Batch: 21/71, Loss: 0.4343, Time: 4.65s
Epoch: 8, Batch: 31/71, Loss: 0.4120, Time: 4.54s
Epoch: 8, Batch: 41/71, Loss: 0.3237, Time: 6.29s
Epoch: 8, Batch: 51/71, Loss: 0.3678, Time: 5.09s
Epoch: 8, Batch: 61/71, Loss: 0.2876, Time: 4.91s
Epoch: 8, Batch: 71/71, Loss: 0.3969, Time: 5.81s
Epoch 8/10: Train Loss: 0.3818, Val Loss: 0.3286, Val IoU: 0.6147, Val Dice: 0.70
20
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Epoch: 9, Batch: 1/71, Loss: 0.3137, Time: 0.48s
Epoch: 9, Batch: 11/71, Loss: 0.4076, Time: 4.47s
Epoch: 9, Batch: 21/71, Loss: 0.3990, Time: 5.56s
Epoch: 9, Batch: 31/71, Loss: 0.2915, Time: 6.56s
Epoch: 9, Batch: 41/71, Loss: 0.3584, Time: 6.03s
Epoch: 9, Batch: 51/71, Loss: 0.4380, Time: 5.43s
Epoch: 9, Batch: 61/71, Loss: 0.3935, Time: 4.92s
Epoch: 9, Batch: 71/71, Loss: 0.3871, Time: 4.54s
Epoch 9/10: Train Loss: 0.3730, Val Loss: 0.3844, Val IoU: 0.6709, Val Dice: 0.75
42
Epoch: 10, Batch: 1/71, Loss: 0.4335, Time: 0.83s
Epoch: 10, Batch: 11/71, Loss: 0.2766, Time: 5.75s
Epoch: 10, Batch: 21/71, Loss: 0.3836, Time: 4.71s
Epoch: 10, Batch: 31/71, Loss: 0.3647, Time: 4.98s
Epoch: 10, Batch: 41/71, Loss: 0.4501, Time: 6.12s
Epoch: 10, Batch: 51/71, Loss: 0.3531, Time: 5.57s
Epoch: 10, Batch: 61/71, Loss: 0.4450, Time: 5.00s
Epoch: 10, Batch: 71/71, Loss: 0.3960, Time: 4.68s
Epoch 10/10: Train Loss: 0.3668, Val Loss: 0.4070, Val IoU: 0.6663, Val Dice: 0.7
501
Training complete! Best IoU: 0.6847
Models saved to waterbody-dataset\unet_resnet34_frac_50_models
Found 284 image files to process
Processing file 1/284: water_body_1044.jpg
Processing file 2/284: water_body_1064.jpg
Processing file 3/284: water_body_107.jpg
Processing file 4/284: water_body_1070.jpg
Processing file 5/284: water_body_11.jpg
Processing file 6/284: water_body_1133.jpg
Processing file 7/284: water_body_1143.jpg
Processing file 8/284: water_body_1152.jpg
Processing file 9/284: water_body_1170.jpg
Processing file 10/284: water_body_118.jpg
Processing file 11/284: water_body_1195.jpg
Processing file 12/284: water_body_120.jpg
Processing file 13/284: water_body_1201.jpg
Processing file 14/284: water_body_1204.jpg
Processing file 15/284: water_body_1228.jpg
Processing file 16/284: water_body_1235.jpg
Processing file 17/284: water_body_1237.jpg
Processing file 18/284: water_body_1253.jpg
Processing file 19/284: water_body_1260.jpg
Processing file 20/284: water_body_1313.jpg
Processing file 21/284: water_body_1329.jpg
Processing file 22/284: water_body_1334.jpg
Processing file 23/284: water_body_1343.jpg
Processing file 24/284: water_body_1347.jpg
Processing file 25/284: water_body_1369.jpg
Processing file 26/284: water_body_1383.jpg
Processing file 27/284: water_body_1417.jpg
Processing file 28/284: water_body_1422.jpg
Processing file 29/284: water_body_1447.jpg
Processing file 30/284: water_body_1453.jpg
Processing file 31/284: water_body_1459.jpg
Processing file 32/284: water_body_1462.jpg
Processing file 33/284: water_body_1470.jpg
Processing file 34/284: water_body_1482.jpg
Processing file 35/284: water_body_1498.jpg
Processing file 36/284: water_body_1509.jpg
Processing file 37/284: water_body_1510.jpg
```

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Processing file 38/284: water_body_1511.jpg
Processing file 39/284: water_body_1529.jpg
Processing file 40/284: water_body_1532.jpg
Processing file 41/284: water_body_1553.jpg
Processing file 42/284: water_body_1575.jpg
Processing file 43/284: water_body_1582.jpg
Processing file 44/284: water_body_1594.jpg
Processing file 45/284: water_body_1597.jpg
Processing file 46/284: water_body_1611.jpg
Processing file 47/284: water_body_1615.jpg
Processing file 48/284: water_body_1618.jpg
Processing file 49/284: water_body_1620.jpg
Processing file 50/284: water_body_1628.jpg
Processing file 51/284: water_body_1634.jpg
Processing file 52/284: water_body_1645.jpg
Processing file 53/284: water_body_1656.jpg
Processing file 54/284: water_body_1657.jpg
Processing file 55/284: water_body_1674.jpg
Processing file 56/284: water_body_1688.jpg
Processing file 57/284: water_body_1691.jpg
Processing file 58/284: water_body_1694.jpg
Processing file 59/284: water_body_1699.jpg
Processing file 60/284: water_body_1703.jpg
Processing file 61/284: water_body_1704.jpg
Processing file 62/284: water_body_1711.jpg
Processing file 63/284: water_body_1724.jpg
Processing file 64/284: water_body_1734.jpg
Processing file 65/284: water_body_1745.jpg
Processing file 66/284: water_body_1749.jpg
Processing file 67/284: water_body_1770.jpg
Processing file 68/284: water_body_1771.jpg
Processing file 69/284: water_body_1776.jpg
Processing file 70/284: water_body_1794.jpg
Processing file 71/284: water_body_1816.jpg
Processing file 72/284: water_body_1832.jpg
Processing file 73/284: water_body_1833.jpg
Processing file 74/284: water_body_1839.jpg
Processing file 75/284: water_body_186.jpg
Processing file 76/284: water_body_2182.jpg
Processing file 77/284: water_body_2203.jpg
Processing file 78/284: water_body_2220.jpg
Processing file 79/284: water_body_2227.jpg
Processing file 80/284: water_body_2235.jpg
Processing file 81/284: water_body_2246.jpg
Processing file 82/284: water_body_2250.jpg
Processing file 83/284: water_body_2253.jpg
Processing file 84/284: water_body_2276.jpg
Processing file 85/284: water_body_2287.jpg
Processing file 86/284: water_body_2291.jpg
Processing file 87/284: water_body_2340.jpg
Processing file 88/284: water_body_2370.jpg
Processing file 89/284: water_body_2372.jpg
Processing file 90/284: water_body_2373.jpg
Processing file 91/284: water_body_2395.jpg
Processing file 92/284: water_body_2397.jpg
Processing file 93/284: water_body_2415.jpg
Processing file 94/284: water_body_2432.jpg
Processing file 95/284: water_body_2440.jpg
Processing file 96/284: water_body_2456.jpg
Processing file 97/284: water_body_2500.jpg
```

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Processing file 98/284: water_body_2504.jpg
Processing file 99/284: water_body_2527.jpg
Processing file 100/284: water_body_2530.jpg
Processing file 101/284: water_body_2532.jpg
Processing file 102/284: water_body_2533.jpg
Processing file 103/284: water_body_2554.jpg
Processing file 104/284: water_body_2561.jpg
Processing file 105/284: water_body_29.jpg
Processing file 106/284: water_body_42.jpg
Processing file 107/284: water_body_59.jpg
Processing file 108/284: water_body_606.jpg
Processing file 109/284: water_body_631.jpg
Processing file 110/284: water_body_6378.jpg
Processing file 111/284: water_body_6389.jpg
Processing file 112/284: water_body_6421.jpg
Processing file 113/284: water_body_644.jpg
Processing file 114/284: water_body_65.jpg
Processing file 115/284: water_body_654.jpg
Processing file 116/284: water_body_7037.jpg
Processing file 117/284: water_body_7045.jpg
Processing file 118/284: water_body_7056.jpg
Processing file 119/284: water_body_7058.jpg
Processing file 120/284: water_body_7071.jpg
Processing file 121/284: water_body_7090.jpg
Processing file 122/284: water_body_7095.jpg
Processing file 123/284: water_body_71.jpg
Processing file 124/284: water_body_7137.jpg
Processing file 125/284: water_body_7140.jpg
Processing file 126/284: water_body_7147.jpg
Processing file 127/284: water_body_7162.jpg
Processing file 128/284: water_body_7165.jpg
Processing file 129/284: water_body_7172.jpg
Processing file 130/284: water_body_718.jpg
Processing file 131/284: water_body_7184.jpg
Processing file 132/284: water_body_7192.jpg
Processing file 133/284: water_body_720.jpg
Processing file 134/284: water_body_7205.jpg
Processing file 135/284: water_body_7228.jpg
Processing file 136/284: water_body_7230.jpg
Processing file 137/284: water_body_7232.jpg
Processing file 138/284: water_body_7244.jpg
Processing file 139/284: water_body_7258.jpg
Processing file 140/284: water_body_7260.jpg
Processing file 141/284: water_body_7262.jpg
Processing file 142/284: water_body_7283.jpg
Processing file 143/284: water_body_7319.jpg
Processing file 144/284: water_body_7333.jpg
Processing file 145/284: water_body_7352.jpg
Processing file 146/284: water_body_7404.jpg
Processing file 147/284: water_body_7420.jpg
Processing file 148/284: water_body_7429.jpg
Processing file 149/284: water_body_7432.jpg
Processing file 150/284: water_body_7436.jpg
Processing file 151/284: water_body_7437.jpg
Processing file 152/284: water_body_7444.jpg
Processing file 153/284: water_body_7475.jpg
Processing file 154/284: water_body_7508.jpg
Processing file 155/284: water_body_7515.jpg
Processing file 156/284: water_body_7517.jpg
Processing file 157/284: water_body_7518.jpg
```

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Processing file 158/284: water_body_7525.jpg
Processing file 159/284: water_body_7540.jpg
Processing file 160/284: water_body_7544.jpg
Processing file 161/284: water_body_7569.jpg
Processing file 162/284: water_body_7626.jpg
Processing file 163/284: water_body_7686.jpg
Processing file 164/284: water_body_7693.jpg
Processing file 165/284: water_body_7696.jpg
Processing file 166/284: water_body_7703.jpg
Processing file 167/284: water_body_7707.jpg
Processing file 168/284: water_body_7708.jpg
Processing file 169/284: water_body_7709.jpg
Processing file 170/284: water_body_7711.jpg
Processing file 171/284: water_body_772.jpg
Processing file 172/284: water_body_7720.jpg
Processing file 173/284: water_body_7727.jpg
Processing file 174/284: water_body_7729.jpg
Processing file 175/284: water_body_773.jpg
Processing file 176/284: water_body_7750.jpg
Processing file 177/284: water_body_7773.jpg
Processing file 178/284: water_body_7781.jpg
Processing file 179/284: water_body_7791.jpg
Processing file 180/284: water_body_7815.jpg
Processing file 181/284: water_body_7823.jpg
Processing file 182/284: water_body_7828.jpg
Processing file 183/284: water_body_783.jpg
Processing file 184/284: water_body_7848.jpg
Processing file 185/284: water_body_7868.jpg
Processing file 186/284: water_body_7869.jpg
Processing file 187/284: water_body_7892.jpg
Processing file 188/284: water_body_7903.jpg
Processing file 189/284: water_body_7944.jpg
Processing file 190/284: water_body_7946.jpg
Processing file 191/284: water_body_7947.jpg
Processing file 192/284: water_body_7954.jpg
Processing file 193/284: water_body_7955.jpg
Processing file 194/284: water_body_7957.jpg
Processing file 195/284: water_body_7978.jpg
Processing file 196/284: water_body_8002.jpg
Processing file 197/284: water_body_8013.jpg
Processing file 198/284: water_body_8017.jpg
Processing file 199/284: water_body_8027.jpg
Processing file 200/284: water_body_8059.jpg
Processing file 201/284: water_body_806.jpg
Processing file 202/284: water_body_8068.jpg
Processing file 203/284: water_body_8076.jpg
Processing file 204/284: water_body_8098.jpg
Processing file 205/284: water_body_810.jpg
Processing file 206/284: water_body_811.jpg
Processing file 207/284: water_body_8165.jpg
Processing file 208/284: water_body_8177.jpg
Processing file 209/284: water_body_818.jpg
Processing file 210/284: water_body_8185.jpg
Processing file 211/284: water_body_8219.jpg
Processing file 212/284: water_body_823.jpg
Processing file 213/284: water_body_8242.jpg
Processing file 214/284: water_body_8248.jpg
Processing file 215/284: water_body_8253.jpg
Processing file 216/284: water_body_8319.jpg
Processing file 217/284: water_body_8338.jpg
```

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Processing file 218/284: water_body_8344.jpg
Processing file 219/284: water_body_8348.jpg
Processing file 220/284: water_body_8356.jpg
Processing file 221/284: water_body_8377.jpg
Processing file 222/284: water_body_8387.jpg
Processing file 223/284: water_body_839.jpg
Processing file 224/284: water_body_8398.jpg
Processing file 225/284: water_body_8408.jpg
Processing file 226/284: water_body_8412.jpg
Processing file 227/284: water_body_8423.jpg
Processing file 228/284: water_body_8424.jpg
Processing file 229/284: water_body_8439.jpg
Processing file 230/284: water_body_8442.jpg
Processing file 231/284: water_body_8455.jpg
Processing file 232/284: water_body_8462.jpg
Processing file 233/284: water_body_847.jpg
Processing file 234/284: water_body_8476.jpg
Processing file 235/284: water_body_8497.jpg
Processing file 236/284: water_body_8558.jpg
Processing file 237/284: water_body_8560.jpg
Processing file 238/284: water_body_8583.jpg
Processing file 239/284: water_body_8607.jpg
Processing file 240/284: water_body_8619.jpg
Processing file 241/284: water_body_8622.jpg
Processing file 242/284: water_body_8633.jpg
Processing file 243/284: water_body_8650.jpg
Processing file 244/284: water_body_8651.jpg
Processing file 245/284: water_body_8653.jpg
Processing file 246/284: water_body_8664.jpg
Processing file 247/284: water_body_8670.jpg
Processing file 248/284: water_body_8671.jpg
Processing file 249/284: water_body_8680.jpg
Processing file 250/284: water_body_8687.jpg
Processing file 251/284: water_body_8693.jpg
Processing file 252/284: water_body_8702.jpg
Processing file 253/284: water_body_8703.jpg
Processing file 254/284: water_body_8715.jpg
Processing file 255/284: water_body_8722.jpg
Processing file 256/284: water_body_8729.jpg
Processing file 257/284: water_body_875.jpg
Processing file 258/284: water_body_8760.jpg
Processing file 259/284: water_body_8763.jpg
Processing file 260/284: water_body_8764.jpg
Processing file 261/284: water_body_8789.jpg
Processing file 262/284: water_body_8823.jpg
Processing file 263/284: water_body_8825.jpg
Processing file 264/284: water_body_8843.jpg
Processing file 265/284: water_body_885.jpg
Processing file 266/284: water_body_8854.jpg
Processing file 267/284: water_body_8877.jpg
Processing file 268/284: water_body_8885.jpg
Processing file 269/284: water_body_8911.jpg
Processing file 270/284: water_body_8918.jpg
Processing file 271/284: water_body_8930.jpg
Processing file 272/284: water_body_8949.jpg
Processing file 273/284: water_body_8953.jpg
Processing file 274/284: water_body_8960.jpg
Processing file 275/284: water_body_897.jpg
Processing file 276/284: water_body_900.jpg
Processing file 277/284: water_body_938.jpg
```

```
Processing file 278/284: water_body_957.jpg
Processing file 279/284: water_body_967.jpg
Processing file 280/284: water_body_970.jpg
Processing file 281/284: water_body_978.jpg
Processing file 282/284: water_body_979.jpg
Processing file 283/284: water_body_987.jpg
Processing file 284/284: water_body_99.jpg
Batch processing completed. Results saved to waterbody-dataset\pred_val_frac_50
[eval] matched 284/284 files in waterbody-dataset\pred_val_frac_50
Found 284 image files to process
Processing file 1/284: water_body_1003.jpg
Processing file 2/284: water_body_1015.jpg
Processing file 3/284: water_body_105.jpg
Processing file 4/284: water_body_1052.jpg
Processing file 5/284: water_body_1059.jpg
Processing file 6/284: water_body_1062.jpg
Processing file 7/284: water_body_1074.jpg
Processing file 8/284: water_body_1076.jpg
Processing file 9/284: water_body_1089.jpg
Processing file 10/284: water_body_1096.jpg
Processing file 11/284: water_body_1104.jpg
Processing file 12/284: water_body_1109.jpg
Processing file 13/284: water_body_1112.jpg
Processing file 14/284: water_body_1124.jpg
Processing file 15/284: water_body_1184.jpg
Processing file 16/284: water_body_1203.jpg
Processing file 17/284: water_body_121.jpg
Processing file 18/284: water_body_1227.jpg
Processing file 19/284: water_body_1280.jpg
Processing file 20/284: water_body_1307.jpg
Processing file 21/284: water_body_1354.jpg
Processing file 22/284: water_body_1359.jpg
Processing file 23/284: water_body_1361.jpg
Processing file 24/284: water_body_1368.jpg
Processing file 25/284: water_body_1373.jpg
Processing file 26/284: water_body_1382.jpg
Processing file 27/284: water_body_1392.jpg
Processing file 28/284: water_body_1406.jpg
Processing file 29/284: water_body_1420.jpg
Processing file 30/284: water_body_1433.jpg
Processing file 31/284: water_body_1439.jpg
Processing file 32/284: water_body_1451.jpg
Processing file 33/284: water_body_1468.jpg
Processing file 34/284: water_body_1502.jpg
Processing file 35/284: water_body_1513.jpg
Processing file 36/284: water_body_1525.jpg
Processing file 37/284: water_body_1559.jpg
Processing file 38/284: water_body_1565.jpg
Processing file 39/284: water_body_1572.jpg
Processing file 40/284: water_body_1577.jpg
Processing file 41/284: water_body_1579.jpg
Processing file 42/284: water_body_1586.jpg
Processing file 43/284: water_body_1591.jpg
Processing file 44/284: water_body_1593.jpg
Processing file 45/284: water_body_1622.jpg
Processing file 46/284: water_body_1623.jpg
Processing file 47/284: water_body_1626.jpg
Processing file 48/284: water_body_1630.jpg
Processing file 49/284: water_body_1655.jpg
Processing file 50/284: water_body_1658.jpg
```

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Processing file 51/284: water_body_1662.jpg
Processing file 52/284: water_body_1663.jpg
Processing file 53/284: water_body_1664.jpg
Processing file 54/284: water_body_1684.jpg
Processing file 55/284: water_body_1693.jpg
Processing file 56/284: water_body_1733.jpg
Processing file 57/284: water_body_1760.jpg
Processing file 58/284: water_body_1768.jpg
Processing file 59/284: water_body_1769.jpg
Processing file 60/284: water_body_1787.jpg
Processing file 61/284: water_body_1796.jpg
Processing file 62/284: water_body_1801.jpg
Processing file 63/284: water_body_1813.jpg
Processing file 64/284: water_body_1815.jpg
Processing file 65/284: water_body_1831.jpg
Processing file 66/284: water_body_1844.jpg
Processing file 67/284: water_body_1847.jpg
Processing file 68/284: water_body_21.jpg
Processing file 69/284: water_body_2171.jpg
Processing file 70/284: water_body_2185.jpg
Processing file 71/284: water_body_2190.jpg
Processing file 72/284: water_body_22.jpg
Processing file 73/284: water_body_2217.jpg
Processing file 74/284: water_body_2255.jpg
Processing file 75/284: water_body_2264.jpg
Processing file 76/284: water_body_2270.jpg
Processing file 77/284: water_body_2282.jpg
Processing file 78/284: water_body_2285.jpg
Processing file 79/284: water_body_2293.jpg
Processing file 80/284: water_body_2295.jpg
Processing file 81/284: water_body_2299.jpg
Processing file 82/284: water_body_2307.jpg
Processing file 83/284: water_body_232.jpg
Processing file 84/284: water_body_2320.jpg
Processing file 85/284: water_body_2333.jpg
Processing file 86/284: water_body_2385.jpg
Processing file 87/284: water_body_2402.jpg
Processing file 88/284: water_body_2472.jpg
Processing file 89/284: water_body_2474.jpg
Processing file 90/284: water_body_2498.jpg
Processing file 91/284: water_body_2520.jpg
Processing file 92/284: water_body_2529.jpg
Processing file 93/284: water_body_2538.jpg
Processing file 94/284: water_body_2541.jpg
Processing file 95/284: water_body_2553.jpg
Processing file 96/284: water_body_2559.jpg
Processing file 97/284: water_body_27.jpg
Processing file 98/284: water_body_39.jpg
Processing file 99/284: water_body_55.jpg
Processing file 100/284: water_body_590.jpg
Processing file 101/284: water_body_599.jpg
Processing file 102/284: water_body_607.jpg
Processing file 103/284: water_body_615.jpg
Processing file 104/284: water_body_620.jpg
Processing file 105/284: water_body_6377.jpg
Processing file 106/284: water_body_6385.jpg
Processing file 107/284: water_body_64.jpg
Processing file 108/284: water_body_6426.jpg
Processing file 109/284: water_body_668.jpg
Processing file 110/284: water_body_68.jpg
```

```
Processing file 111/284: water_body_690.jpg
Processing file 112/284: water_body_693.jpg
Processing file 113/284: water_body_7029.jpg
Processing file 114/284: water_body_704.jpg
Processing file 115/284: water_body_7040.jpg
Processing file 116/284: water_body_7052.jpg
Processing file 117/284: water_body_7057.jpg
Processing file 118/284: water_body_7067.jpg
Processing file 119/284: water_body_7068.jpg
Processing file 120/284: water_body_7079.jpg
Processing file 121/284: water_body_7100.jpg
Processing file 122/284: water_body_7132.jpg
Processing file 123/284: water_body_714.jpg
Processing file 124/284: water_body_7166.jpg
Processing file 125/284: water_body_7168.jpg
Processing file 126/284: water_body_7173.jpg
Processing file 127/284: water_body_7181.jpg
Processing file 128/284: water_body_7191.jpg
Processing file 129/284: water_body_7199.jpg
Processing file 130/284: water_body_72.jpg
Processing file 131/284: water_body_7207.jpg
Processing file 132/284: water_body_7235.jpg
Processing file 133/284: water_body_7238.jpg
Processing file 134/284: water_body_7245.jpg
Processing file 135/284: water_body_7247.jpg
Processing file 136/284: water_body_7252.jpg
Processing file 137/284: water_body_7265.jpg
Processing file 138/284: water_body_7275.jpg
Processing file 139/284: water_body_7276.jpg
Processing file 140/284: water_body_7278.jpg
Processing file 141/284: water_body_7289.jpg
Processing file 142/284: water_body_7290.jpg
Processing file 143/284: water_body_7296.jpg
Processing file 144/284: water_body_7305.jpg
Processing file 145/284: water_body_7327.jpg
Processing file 146/284: water_body_7386.jpg
Processing file 147/284: water_body_7392.jpg
Processing file 148/284: water_body_7395.jpg
Processing file 149/284: water_body_7396.jpg
Processing file 150/284: water_body_7401.jpg
Processing file 151/284: water_body_7413.jpg
Processing file 152/284: water_body_7416.jpg
Processing file 153/284: water_body_7417.jpg
Processing file 154/284: water_body_7418.jpg
Processing file 155/284: water_body_742.jpg
Processing file 156/284: water_body_7421.jpg
Processing file 157/284: water_body_7423.jpg
Processing file 158/284: water_body_7433.jpg
Processing file 159/284: water_body_7438.jpg
Processing file 160/284: water_body_744.jpg
Processing file 161/284: water_body_7486.jpg
Processing file 162/284: water_body_7499.jpg
Processing file 163/284: water_body_7510.jpg
Processing file 164/284: water_body_7511.jpg
Processing file 165/284: water_body_7543.jpg
Processing file 166/284: water_body_7552.jpg
Processing file 167/284: water_body_7562.jpg
Processing file 168/284: water_body_7575.jpg
Processing file 169/284: water_body_7578.jpg
Processing file 170/284: water_body_7581.jpg
```

```
Processing file 171/284: water_body_7586.jpg
Processing file 172/284: water_body_759.jpg
Processing file 173/284: water_body_7603.jpg
Processing file 174/284: water_body_7615.jpg
Processing file 175/284: water_body_7618.jpg
Processing file 176/284: water_body_7629.jpg
Processing file 177/284: water_body_7638.jpg
Processing file 178/284: water_body_7673.jpg
Processing file 179/284: water_body_7701.jpg
Processing file 180/284: water_body_771.jpg
Processing file 181/284: water_body_7712.jpg
Processing file 182/284: water_body_7717.jpg
Processing file 183/284: water_body_7766.jpg
Processing file 184/284: water_body_7793.jpg
Processing file 185/284: water_body_78.jpg
Processing file 186/284: water_body_7810.jpg
Processing file 187/284: water_body_7843.jpg
Processing file 188/284: water_body_7871.jpg
Processing file 189/284: water_body_788.jpg
Processing file 190/284: water_body_7891.jpg
Processing file 191/284: water_body_790.jpg
Processing file 192/284: water_body_7901.jpg
Processing file 193/284: water_body_7914.jpg
Processing file 194/284: water_body_7918.jpg
Processing file 195/284: water_body_7956.jpg
Processing file 196/284: water_body_796.jpg
Processing file 197/284: water_body_7961.jpg
Processing file 198/284: water_body_7975.jpg
Processing file 199/284: water_body_7977.jpg
Processing file 200/284: water_body_800.jpg
Processing file 201/284: water_body_8005.jpg
Processing file 202/284: water_body_801.jpg
Processing file 203/284: water_body_8018.jpg
Processing file 204/284: water_body_8033.jpg
Processing file 205/284: water_body_8052.jpg
Processing file 206/284: water_body_8064.jpg
Processing file 207/284: water_body_8140.jpg
Processing file 208/284: water_body_8146.jpg
Processing file 209/284: water_body_8156.jpg
Processing file 210/284: water_body_8159.jpg
Processing file 211/284: water_body_8169.jpg
Processing file 212/284: water_body_8195.jpg
Processing file 213/284: water_body_8206.jpg
Processing file 214/284: water_body_8245.jpg
Processing file 215/284: water_body_825.jpg
Processing file 216/284: water_body_8254.jpg
Processing file 217/284: water_body_8273.jpg
Processing file 218/284: water_body_8288.jpg
Processing file 219/284: water_body_8321.jpg
Processing file 220/284: water_body_8352.jpg
Processing file 221/284: water_body_8367.jpg
Processing file 222/284: water_body_8376.jpg
Processing file 223/284: water_body_838.jpg
Processing file 224/284: water_body_8388.jpg
Processing file 225/284: water_body_8402.jpg
Processing file 226/284: water_body_8405.jpg
Processing file 227/284: water_body_841.jpg
Processing file 228/284: water_body_8446.jpg
Processing file 229/284: water_body_8448.jpg
Processing file 230/284: water_body_8463.jpg
```

```
Processing file 231/284: water_body_8467.jpg
Processing file 232/284: water_body_8482.jpg
Processing file 233/284: water_body_8491.jpg
Processing file 234/284: water_body_8495.jpg
Processing file 235/284: water_body_8501.jpg
Processing file 236/284: water_body_8504.jpg
Processing file 237/284: water_body_8506.jpg
Processing file 238/284: water_body_8509.jpg
Processing file 239/284: water_body_851.jpg
Processing file 240/284: water_body_8526.jpg
Processing file 241/284: water_body_8533.jpg
Processing file 242/284: water_body_8546.jpg
Processing file 243/284: water_body_8556.jpg
Processing file 244/284: water_body_8561.jpg
Processing file 245/284: water_body_8573.jpg
Processing file 246/284: water_body_8584.jpg
Processing file 247/284: water_body_8616.jpg
Processing file 248/284: water_body_8632.jpg
Processing file 249/284: water_body_8636.jpg
Processing file 250/284: water_body_8641.jpg
Processing file 251/284: water_body_8655.jpg
Processing file 252/284: water_body_8659.jpg
Processing file 253/284: water_body_8676.jpg
Processing file 254/284: water_body_8678.jpg
Processing file 255/284: water_body_8679.jpg
Processing file 256/284: water_body_8694.jpg
Processing file 257/284: water_body_8717.jpg
Processing file 258/284: water_body_8728.jpg
Processing file 259/284: water_body_8730.jpg
Processing file 260/284: water_body_8738.jpg
Processing file 261/284: water_body_8749.jpg
Processing file 262/284: water_body_8752.jpg
Processing file 263/284: water_body_8753.jpg
Processing file 264/284: water_body_8770.jpg
Processing file 265/284: water_body_8774.jpg
Processing file 266/284: water_body_8787.jpg
Processing file 267/284: water_body_8792.jpg
Processing file 268/284: water_body_881.jpg
Processing file 269/284: water_body_8847.jpg
Processing file 270/284: water_body_8850.jpg
Processing file 271/284: water_body_8851.jpg
Processing file 272/284: water_body_8859.jpg
Processing file 273/284: water_body_889.jpg
Processing file 274/284: water_body_8912.jpg
Processing file 275/284: water_body_8921.jpg
Processing file 276/284: water_body_8926.jpg
Processing file 277/284: water_body_8934.jpg
Processing file 278/284: water_body_8935.jpg
Processing file 279/284: water_body_918.jpg
Processing file 280/284: water_body_949.jpg
Processing file 281/284: water_body_96.jpg
Processing file 282/284: water_body_964.jpg
Processing file 283/284: water_body_968.jpg
Processing file 284/284: water_body_994.jpg
Batch processing completed. Results saved to waterbody-dataset\pred_test_frac_50
[eval] matched 284/284 files in waterbody-dataset\pred_test_frac_50
VAL  -> OA:0.7732 | IoU:0.3207 | Dice:0.3897
TEST -> OA:0.7635 | IoU:0.3381 | Dice:0.4097
Model params: total=24,436,514 | trainable=24,436,514 | size=93.38 MB
Train time: 378.7s | Inference: 223.5 ms/img
```

```
=== Fraction 100% ===
Training subset size: 2273
Internal val_split used: 0.0040
Using device: cuda
Found 2273 image files and 2273 label files
Training on 2263 images, validating on 10 images
Testing data loader...
Data loader test passed.
Starting training with unet + resnet34
Model parameters: 24,436,514
Epoch: 1, Batch: 1/142, Loss: 0.7437, Time: 0.60s
Epoch: 1, Batch: 11/142, Loss: 0.5073, Time: 4.65s
Epoch: 1, Batch: 21/142, Loss: 0.5199, Time: 5.79s
Epoch: 1, Batch: 31/142, Loss: 0.4169, Time: 5.38s
Epoch: 1, Batch: 41/142, Loss: 0.3565, Time: 4.83s
Epoch: 1, Batch: 51/142, Loss: 0.8266, Time: 5.55s
Epoch: 1, Batch: 61/142, Loss: 0.4219, Time: 5.25s
Epoch: 1, Batch: 71/142, Loss: 0.3904, Time: 4.70s
Epoch: 1, Batch: 81/142, Loss: 0.3949, Time: 5.51s
Epoch: 1, Batch: 91/142, Loss: 0.4467, Time: 4.82s
Epoch: 1, Batch: 101/142, Loss: 0.3666, Time: 5.51s
Epoch: 1, Batch: 111/142, Loss: 0.3698, Time: 5.32s
Epoch: 1, Batch: 121/142, Loss: 0.3864, Time: 5.15s
Epoch: 1, Batch: 131/142, Loss: 0.3694, Time: 5.66s
Epoch: 1, Batch: 141/142, Loss: 0.3994, Time: 7.23s
Epoch 1/10: Train Loss: 0.4560, Val Loss: 0.5841, Val IoU: 0.6016, Val Dice: 0.70
30
Saving best model with IoU: 0.6016
Epoch: 2, Batch: 1/142, Loss: 0.3506, Time: 0.42s
Epoch: 2, Batch: 11/142, Loss: 0.4048, Time: 5.55s
Epoch: 2, Batch: 21/142, Loss: 0.3948, Time: 4.98s
Epoch: 2, Batch: 31/142, Loss: 0.4102, Time: 4.58s
Epoch: 2, Batch: 41/142, Loss: 0.3160, Time: 5.02s
Epoch: 2, Batch: 51/142, Loss: 0.3779, Time: 6.05s
Epoch: 2, Batch: 61/142, Loss: 0.4856, Time: 4.84s
Epoch: 2, Batch: 71/142, Loss: 0.4682, Time: 5.89s
Epoch: 2, Batch: 81/142, Loss: 0.4987, Time: 5.23s
Epoch: 2, Batch: 91/142, Loss: 0.3348, Time: 5.54s
Epoch: 2, Batch: 101/142, Loss: 0.4355, Time: 4.98s
Epoch: 2, Batch: 111/142, Loss: 0.4285, Time: 5.66s
Epoch: 2, Batch: 121/142, Loss: 0.3201, Time: 5.31s
Epoch: 2, Batch: 131/142, Loss: 0.3122, Time: 5.17s
Epoch: 2, Batch: 141/142, Loss: 0.5840, Time: 5.83s
Epoch 2/10: Train Loss: 0.4130, Val Loss: 0.5027, Val IoU: 0.6039, Val Dice: 0.70
94
Saving best model with IoU: 0.6039
Epoch: 3, Batch: 1/142, Loss: 0.4787, Time: 0.35s
Epoch: 3, Batch: 11/142, Loss: 0.3953, Time: 4.73s
Epoch: 3, Batch: 21/142, Loss: 0.4169, Time: 6.32s
Epoch: 3, Batch: 31/142, Loss: 0.3345, Time: 4.63s
Epoch: 3, Batch: 41/142, Loss: 0.3147, Time: 5.69s
Epoch: 3, Batch: 51/142, Loss: 0.5117, Time: 5.94s
Epoch: 3, Batch: 61/142, Loss: 0.3506, Time: 5.48s
Epoch: 3, Batch: 71/142, Loss: 0.3780, Time: 4.34s
Epoch: 3, Batch: 81/142, Loss: 0.2825, Time: 4.76s
Epoch: 3, Batch: 91/142, Loss: 0.3494, Time: 5.41s
Epoch: 3, Batch: 101/142, Loss: 0.2906, Time: 4.57s
Epoch: 3, Batch: 111/142, Loss: 0.3378, Time: 5.49s
Epoch: 3, Batch: 121/142, Loss: 0.4057, Time: 6.07s
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Epoch: 3, Batch: 131/142, Loss: 0.4172, Time: 5.86s
Epoch: 3, Batch: 141/142, Loss: 0.3003, Time: 5.61s
Epoch 3/10: Train Loss: 0.3837, Val Loss: 0.5972, Val IoU: 0.5792, Val Dice: 0.68
19
Epoch: 4, Batch: 1/142, Loss: 0.3440, Time: 0.58s
Epoch: 4, Batch: 11/142, Loss: 0.3753, Time: 4.83s
Epoch: 4, Batch: 21/142, Loss: 0.3292, Time: 5.44s
Epoch: 4, Batch: 31/142, Loss: 0.4070, Time: 4.72s
Epoch: 4, Batch: 41/142, Loss: 0.5424, Time: 4.97s
Epoch: 4, Batch: 51/142, Loss: 0.3966, Time: 4.97s
Epoch: 4, Batch: 61/142, Loss: 0.3795, Time: 7.37s
Epoch: 4, Batch: 71/142, Loss: 0.3708, Time: 4.62s
Epoch: 4, Batch: 81/142, Loss: 0.4833, Time: 5.39s
Epoch: 4, Batch: 91/142, Loss: 0.3644, Time: 5.73s
Epoch: 4, Batch: 101/142, Loss: 0.3535, Time: 4.87s
Epoch: 4, Batch: 111/142, Loss: 0.4079, Time: 5.37s
Epoch: 4, Batch: 121/142, Loss: 0.3082, Time: 5.19s
Epoch: 4, Batch: 131/142, Loss: 0.2784, Time: 5.67s
Epoch: 4, Batch: 141/142, Loss: 0.3444, Time: 5.54s
Epoch 4/10: Train Loss: 0.3809, Val Loss: 0.5929, Val IoU: 0.6305, Val Dice: 0.71
83
Saving best model with IoU: 0.6305
Epoch: 5, Batch: 1/142, Loss: 0.3806, Time: 0.35s
Epoch: 5, Batch: 11/142, Loss: 0.3101, Time: 5.28s
Epoch: 5, Batch: 21/142, Loss: 0.2889, Time: 4.87s
Epoch: 5, Batch: 31/142, Loss: 0.3632, Time: 5.03s
Epoch: 5, Batch: 41/142, Loss: 0.2821, Time: 6.13s
Epoch: 5, Batch: 51/142, Loss: 0.3879, Time: 4.97s
Epoch: 5, Batch: 61/142, Loss: 0.3126, Time: 4.99s
Epoch: 5, Batch: 71/142, Loss: 0.3508, Time: 5.20s
Epoch: 5, Batch: 81/142, Loss: 0.3065, Time: 6.29s
Epoch: 5, Batch: 91/142, Loss: 0.3341, Time: 5.68s
Epoch: 5, Batch: 101/142, Loss: 0.4373, Time: 5.40s
Epoch: 5, Batch: 111/142, Loss: 0.2917, Time: 5.45s
Epoch: 5, Batch: 121/142, Loss: 0.4207, Time: 5.41s
Epoch: 5, Batch: 131/142, Loss: 0.3464, Time: 5.14s
Epoch: 5, Batch: 141/142, Loss: 0.2396, Time: 5.11s
Epoch 5/10: Train Loss: 0.3686, Val Loss: 0.5386, Val IoU: 0.6407, Val Dice: 0.72
44
Saving best model with IoU: 0.6407
Epoch: 6, Batch: 1/142, Loss: 0.3267, Time: 0.41s
Epoch: 6, Batch: 11/142, Loss: 0.3668, Time: 5.58s
Epoch: 6, Batch: 21/142, Loss: 0.3120, Time: 6.51s
Epoch: 6, Batch: 31/142, Loss: 0.2814, Time: 5.81s
Epoch: 6, Batch: 41/142, Loss: 0.3836, Time: 5.09s
Epoch: 6, Batch: 51/142, Loss: 0.2909, Time: 5.66s
Epoch: 6, Batch: 61/142, Loss: 0.2752, Time: 4.36s
Epoch: 6, Batch: 71/142, Loss: 0.3030, Time: 4.85s
Epoch: 6, Batch: 81/142, Loss: 0.4745, Time: 5.35s
Epoch: 6, Batch: 91/142, Loss: 0.4487, Time: 4.77s
Epoch: 6, Batch: 101/142, Loss: 0.3694, Time: 4.91s
Epoch: 6, Batch: 111/142, Loss: 0.3115, Time: 6.38s
Epoch: 6, Batch: 121/142, Loss: 0.3964, Time: 5.22s
Epoch: 6, Batch: 131/142, Loss: 0.3325, Time: 5.69s
Epoch: 6, Batch: 141/142, Loss: 0.3087, Time: 4.87s
Epoch 6/10: Train Loss: 0.3673, Val Loss: 0.6658, Val IoU: 0.5702, Val Dice: 0.66
34
Epoch: 7, Batch: 1/142, Loss: 0.2868, Time: 0.54s
Epoch: 7, Batch: 11/142, Loss: 0.4121, Time: 4.87s
Epoch: 7, Batch: 21/142, Loss: 0.3189, Time: 6.38s
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Epoch: 7, Batch: 31/142, Loss: 0.3481, Time: 4.62s
Epoch: 7, Batch: 41/142, Loss: 0.4193, Time: 5.09s
Epoch: 7, Batch: 51/142, Loss: 0.5163, Time: 4.82s
Epoch: 7, Batch: 61/142, Loss: 0.3876, Time: 4.94s
Epoch: 7, Batch: 71/142, Loss: 0.4853, Time: 5.63s
Epoch: 7, Batch: 81/142, Loss: 0.3121, Time: 6.21s
Epoch: 7, Batch: 91/142, Loss: 0.4262, Time: 4.66s
Epoch: 7, Batch: 101/142, Loss: 0.3416, Time: 5.37s
Epoch: 7, Batch: 111/142, Loss: 0.3405, Time: 4.83s
Epoch: 7, Batch: 121/142, Loss: 0.3834, Time: 4.73s
Epoch: 7, Batch: 131/142, Loss: 0.3177, Time: 6.32s
Epoch: 7, Batch: 141/142, Loss: 0.3024, Time: 6.47s
Epoch 7/10: Train Loss: 0.3572, Val Loss: 0.4336, Val IoU: 0.6385, Val Dice: 0.73
59
Epoch: 8, Batch: 1/142, Loss: 0.3608, Time: 0.36s
Epoch: 8, Batch: 11/142, Loss: 0.4244, Time: 4.40s
Epoch: 8, Batch: 21/142, Loss: 0.2695, Time: 5.28s
Epoch: 8, Batch: 31/142, Loss: 0.4654, Time: 5.17s
Epoch: 8, Batch: 41/142, Loss: 0.3574, Time: 4.04s
Epoch: 8, Batch: 51/142, Loss: 0.3672, Time: 5.80s
Epoch: 8, Batch: 61/142, Loss: 0.5051, Time: 5.55s
Epoch: 8, Batch: 71/142, Loss: 0.4242, Time: 6.59s
Epoch: 8, Batch: 81/142, Loss: 0.4155, Time: 6.21s
Epoch: 8, Batch: 91/142, Loss: 0.3456, Time: 4.83s
Epoch: 8, Batch: 101/142, Loss: 0.3154, Time: 5.71s
Epoch: 8, Batch: 111/142, Loss: 0.3543, Time: 5.81s
Epoch: 8, Batch: 121/142, Loss: 0.3516, Time: 4.12s
Epoch: 8, Batch: 131/142, Loss: 0.3091, Time: 5.82s
Epoch: 8, Batch: 141/142, Loss: 0.3912, Time: 5.50s
Epoch 8/10: Train Loss: 0.3525, Val Loss: 0.5136, Val IoU: 0.6499, Val Dice: 0.73
14
Saving best model with IoU: 0.6499
Epoch: 9, Batch: 1/142, Loss: 0.3259, Time: 0.93s
Epoch: 9, Batch: 11/142, Loss: 0.2535, Time: 4.74s
Epoch: 9, Batch: 21/142, Loss: 0.4297, Time: 7.16s
Epoch: 9, Batch: 31/142, Loss: 0.2813, Time: 4.43s
Epoch: 9, Batch: 41/142, Loss: 0.3413, Time: 4.89s
Epoch: 9, Batch: 51/142, Loss: 0.2983, Time: 5.87s
Epoch: 9, Batch: 61/142, Loss: 0.3629, Time: 4.52s
Epoch: 9, Batch: 71/142, Loss: 0.3468, Time: 5.42s
Epoch: 9, Batch: 81/142, Loss: 0.4287, Time: 5.19s
Epoch: 9, Batch: 91/142, Loss: 0.3371, Time: 6.19s
Epoch: 9, Batch: 101/142, Loss: 0.4114, Time: 4.49s
Epoch: 9, Batch: 111/142, Loss: 0.4825, Time: 4.82s
Epoch: 9, Batch: 121/142, Loss: 0.3848, Time: 5.79s
Epoch: 9, Batch: 131/142, Loss: 0.2986, Time: 5.28s
Epoch: 9, Batch: 141/142, Loss: 0.2875, Time: 5.28s
Epoch 9/10: Train Loss: 0.3505, Val Loss: 0.5239, Val IoU: 0.6322, Val Dice: 0.71
93
Epoch: 10, Batch: 1/142, Loss: 0.2836, Time: 0.46s
Epoch: 10, Batch: 11/142, Loss: 0.3394, Time: 5.20s
Epoch: 10, Batch: 21/142, Loss: 0.3142, Time: 5.44s
Epoch: 10, Batch: 31/142, Loss: 0.3352, Time: 6.42s
Epoch: 10, Batch: 41/142, Loss: 0.4490, Time: 4.87s
Epoch: 10, Batch: 51/142, Loss: 0.3142, Time: 5.28s
Epoch: 10, Batch: 61/142, Loss: 0.3159, Time: 4.62s
Epoch: 10, Batch: 71/142, Loss: 0.3146, Time: 4.32s
Epoch: 10, Batch: 81/142, Loss: 0.5205, Time: 4.86s
Epoch: 10, Batch: 91/142, Loss: 0.2871, Time: 4.87s
Epoch: 10, Batch: 101/142, Loss: 0.3474, Time: 4.70s
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Epoch: 10, Batch: 111/142, Loss: 0.3431, Time: 6.67s
Epoch: 10, Batch: 121/142, Loss: 0.3456, Time: 5.22s
Epoch: 10, Batch: 131/142, Loss: 0.4118, Time: 6.98s
Epoch: 10, Batch: 141/142, Loss: 0.4140, Time: 5.00s
Epoch 10/10: Train Loss: 0.3508, Val Loss: 0.4281, Val IoU: 0.6614, Val Dice: 0.7
558
Saving best model with IoU: 0.6614
Training complete! Best IoU: 0.6614
Models saved to waterbody-dataset\unet_resnet34_frac_100_models
Found 284 image files to process
Processing file 1/284: water_body_1044.jpg
Processing file 2/284: water_body_1064.jpg
Processing file 3/284: water_body_107.jpg
Processing file 4/284: water_body_1070.jpg
Processing file 5/284: water_body_11.jpg
Processing file 6/284: water_body_1133.jpg
Processing file 7/284: water_body_1143.jpg
Processing file 8/284: water_body_1152.jpg
Processing file 9/284: water_body_1170.jpg
Processing file 10/284: water_body_118.jpg
Processing file 11/284: water_body_1195.jpg
Processing file 12/284: water_body_120.jpg
Processing file 13/284: water_body_1201.jpg
Processing file 14/284: water_body_1204.jpg
Processing file 15/284: water_body_1228.jpg
Processing file 16/284: water_body_1235.jpg
Processing file 17/284: water_body_1237.jpg
Processing file 18/284: water_body_1253.jpg
Processing file 19/284: water_body_1260.jpg
Processing file 20/284: water_body_1313.jpg
Processing file 21/284: water_body_1329.jpg
Processing file 22/284: water_body_1334.jpg
Processing file 23/284: water_body_1343.jpg
Processing file 24/284: water_body_1347.jpg
Processing file 25/284: water_body_1369.jpg
Processing file 26/284: water_body_1383.jpg
Processing file 27/284: water_body_1417.jpg
Processing file 28/284: water_body_1422.jpg
Processing file 29/284: water_body_1447.jpg
Processing file 30/284: water_body_1453.jpg
Processing file 31/284: water_body_1459.jpg
Processing file 32/284: water_body_1462.jpg
Processing file 33/284: water_body_1470.jpg
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Processing file 35/284: water_body_1498.jpg
Processing file 36/284: water_body_1509.jpg
Processing file 37/284: water_body_1510.jpg
Processing file 38/284: water_body_1511.jpg
Processing file 39/284: water_body_1529.jpg
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Processing file 41/284: water_body_1553.jpg
Processing file 42/284: water_body_1575.jpg
Processing file 43/284: water_body_1582.jpg
Processing file 44/284: water_body_1594.jpg
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Processing file 46/284: water_body_1611.jpg
Processing file 47/284: water_body_1615.jpg
Processing file 48/284: water_body_1618.jpg
Processing file 49/284: water_body_1620.jpg
Processing file 50/284: water_body_1628.jpg
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Processing file 51/284: water_body_1634.jpg
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Processing file 56/284: water_body_1688.jpg
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Processing file 58/284: water_body_1694.jpg
Processing file 59/284: water_body_1699.jpg
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Processing file 76/284: water_body_2182.jpg
Processing file 77/284: water_body_2203.jpg
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Processing file 103/284: water_body_2554.jpg
Processing file 104/284: water_body_2561.jpg
Processing file 105/284: water_body_29.jpg
Processing file 106/284: water_body_42.jpg
Processing file 107/284: water_body_59.jpg
Processing file 108/284: water_body_606.jpg
Processing file 109/284: water_body_631.jpg
Processing file 110/284: water_body_6378.jpg
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Processing file 111/284: water_body_6389.jpg
Processing file 112/284: water_body_6421.jpg
Processing file 113/284: water_body_644.jpg
Processing file 114/284: water_body_65.jpg
Processing file 115/284: water_body_654.jpg
Processing file 116/284: water_body_7037.jpg
Processing file 117/284: water_body_7045.jpg
Processing file 118/284: water_body_7056.jpg
Processing file 119/284: water_body_7058.jpg
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Processing file 123/284: water_body_71.jpg
Processing file 124/284: water_body_7137.jpg
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Processing file 133/284: water_body_720.jpg
Processing file 134/284: water_body_7205.jpg
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Processing file 170/284: water_body_7711.jpg
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Processing file 205/284: water_body_810.jpg
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Processing file 229/284: water_body_8439.jpg
Processing file 230/284: water_body_8442.jpg
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Processing file 231/284: water_body_8455.jpg
Processing file 232/284: water_body_8462.jpg
Processing file 233/284: water_body_847.jpg
Processing file 234/284: water_body_8476.jpg
Processing file 235/284: water_body_8497.jpg
Processing file 236/284: water_body_8558.jpg
Processing file 237/284: water_body_8560.jpg
Processing file 238/284: water_body_8583.jpg
Processing file 239/284: water_body_8607.jpg
Processing file 240/284: water_body_8619.jpg
Processing file 241/284: water_body_8622.jpg
Processing file 242/284: water_body_8633.jpg
Processing file 243/284: water_body_8650.jpg
Processing file 244/284: water_body_8651.jpg
Processing file 245/284: water_body_8653.jpg
Processing file 246/284: water_body_8664.jpg
Processing file 247/284: water_body_8670.jpg
Processing file 248/284: water_body_8671.jpg
Processing file 249/284: water_body_8680.jpg
Processing file 250/284: water_body_8687.jpg
Processing file 251/284: water_body_8693.jpg
Processing file 252/284: water_body_8702.jpg
Processing file 253/284: water_body_8703.jpg
Processing file 254/284: water_body_8715.jpg
Processing file 255/284: water_body_8722.jpg
Processing file 256/284: water_body_8729.jpg
Processing file 257/284: water_body_875.jpg
Processing file 258/284: water_body_8760.jpg
Processing file 259/284: water_body_8763.jpg
Processing file 260/284: water_body_8764.jpg
Processing file 261/284: water_body_8789.jpg
Processing file 262/284: water_body_8823.jpg
Processing file 263/284: water_body_8825.jpg
Processing file 264/284: water_body_8843.jpg
Processing file 265/284: water_body_885.jpg
Processing file 266/284: water_body_8854.jpg
Processing file 267/284: water_body_8877.jpg
Processing file 268/284: water_body_8885.jpg
Processing file 269/284: water_body_8911.jpg
Processing file 270/284: water_body_8918.jpg
Processing file 271/284: water_body_8930.jpg
Processing file 272/284: water_body_8949.jpg
Processing file 273/284: water_body_8953.jpg
Processing file 274/284: water_body_8960.jpg
Processing file 275/284: water_body_897.jpg
Processing file 276/284: water_body_900.jpg
Processing file 277/284: water_body_938.jpg
Processing file 278/284: water_body_957.jpg
Processing file 279/284: water_body_967.jpg
Processing file 280/284: water_body_970.jpg
Processing file 281/284: water_body_978.jpg
Processing file 282/284: water_body_979.jpg
Processing file 283/284: water_body_987.jpg
Processing file 284/284: water_body_99.jpg
Batch processing completed. Results saved to waterbody-dataset\pred_val_frac_100
[eval] matched 284/284 files in waterbody-dataset\pred_val_frac_100
Found 284 image files to process
Processing file 1/284: water_body_1003.jpg
Processing file 2/284: water_body_1015.jpg
Processing file 3/284: water_body_105.jpg
```

```
Processing file 4/284: water_body_1052.jpg
Processing file 5/284: water_body_1059.jpg
Processing file 6/284: water_body_1062.jpg
Processing file 7/284: water_body_1074.jpg
Processing file 8/284: water_body_1076.jpg
Processing file 9/284: water_body_1089.jpg
Processing file 10/284: water_body_1096.jpg
Processing file 11/284: water_body_1104.jpg
Processing file 12/284: water_body_1109.jpg
Processing file 13/284: water_body_1112.jpg
Processing file 14/284: water_body_1124.jpg
Processing file 15/284: water_body_1184.jpg
Processing file 16/284: water_body_1203.jpg
Processing file 17/284: water_body_121.jpg
Processing file 18/284: water_body_1227.jpg
Processing file 19/284: water_body_1280.jpg
Processing file 20/284: water_body_1307.jpg
Processing file 21/284: water_body_1354.jpg
Processing file 22/284: water_body_1359.jpg
Processing file 23/284: water_body_1361.jpg
Processing file 24/284: water_body_1368.jpg
Processing file 25/284: water_body_1373.jpg
Processing file 26/284: water_body_1382.jpg
Processing file 27/284: water_body_1392.jpg
Processing file 28/284: water_body_1406.jpg
Processing file 29/284: water_body_1420.jpg
Processing file 30/284: water_body_1433.jpg
Processing file 31/284: water_body_1439.jpg
Processing file 32/284: water_body_1451.jpg
Processing file 33/284: water_body_1468.jpg
Processing file 34/284: water_body_1502.jpg
Processing file 35/284: water_body_1513.jpg
Processing file 36/284: water_body_1525.jpg
Processing file 37/284: water_body_1559.jpg
Processing file 38/284: water_body_1565.jpg
Processing file 39/284: water_body_1572.jpg
Processing file 40/284: water_body_1577.jpg
Processing file 41/284: water_body_1579.jpg
Processing file 42/284: water_body_1586.jpg
Processing file 43/284: water_body_1591.jpg
Processing file 44/284: water_body_1593.jpg
Processing file 45/284: water_body_1622.jpg
Processing file 46/284: water_body_1623.jpg
Processing file 47/284: water_body_1626.jpg
Processing file 48/284: water_body_1630.jpg
Processing file 49/284: water_body_1655.jpg
Processing file 50/284: water_body_1658.jpg
Processing file 51/284: water_body_1662.jpg
Processing file 52/284: water_body_1663.jpg
Processing file 53/284: water_body_1664.jpg
Processing file 54/284: water_body_1684.jpg
Processing file 55/284: water_body_1693.jpg
Processing file 56/284: water_body_1733.jpg
Processing file 57/284: water_body_1760.jpg
Processing file 58/284: water_body_1768.jpg
Processing file 59/284: water_body_1769.jpg
Processing file 60/284: water_body_1787.jpg
Processing file 61/284: water_body_1796.jpg
Processing file 62/284: water_body_1801.jpg
Processing file 63/284: water_body_1813.jpg
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Processing file 64/284: water_body_1815.jpg
Processing file 65/284: water_body_1831.jpg
Processing file 66/284: water_body_1844.jpg
Processing file 67/284: water_body_1847.jpg
Processing file 68/284: water_body_21.jpg
Processing file 69/284: water_body_2171.jpg
Processing file 70/284: water_body_2185.jpg
Processing file 71/284: water_body_2190.jpg
Processing file 72/284: water_body_22.jpg
Processing file 73/284: water_body_2217.jpg
Processing file 74/284: water_body_2255.jpg
Processing file 75/284: water_body_2264.jpg
Processing file 76/284: water_body_2270.jpg
Processing file 77/284: water_body_2282.jpg
Processing file 78/284: water_body_2285.jpg
Processing file 79/284: water_body_2293.jpg
Processing file 80/284: water_body_2295.jpg
Processing file 81/284: water_body_2299.jpg
Processing file 82/284: water_body_2307.jpg
Processing file 83/284: water_body_232.jpg
Processing file 84/284: water_body_2320.jpg
Processing file 85/284: water_body_2333.jpg
Processing file 86/284: water_body_2385.jpg
Processing file 87/284: water_body_2402.jpg
Processing file 88/284: water_body_2472.jpg
Processing file 89/284: water_body_2474.jpg
Processing file 90/284: water_body_2498.jpg
Processing file 91/284: water_body_2520.jpg
Processing file 92/284: water_body_2529.jpg
Processing file 93/284: water_body_2538.jpg
Processing file 94/284: water_body_2541.jpg
Processing file 95/284: water_body_2553.jpg
Processing file 96/284: water_body_2559.jpg
Processing file 97/284: water_body_27.jpg
Processing file 98/284: water_body_39.jpg
Processing file 99/284: water_body_55.jpg
Processing file 100/284: water_body_590.jpg
Processing file 101/284: water_body_599.jpg
Processing file 102/284: water_body_607.jpg
Processing file 103/284: water_body_615.jpg
Processing file 104/284: water_body_620.jpg
Processing file 105/284: water_body_6377.jpg
Processing file 106/284: water_body_6385.jpg
Processing file 107/284: water_body_64.jpg
Processing file 108/284: water_body_6426.jpg
Processing file 109/284: water_body_668.jpg
Processing file 110/284: water_body_68.jpg
Processing file 111/284: water_body_690.jpg
Processing file 112/284: water_body_693.jpg
Processing file 113/284: water_body_7029.jpg
Processing file 114/284: water_body_704.jpg
Processing file 115/284: water_body_7040.jpg
Processing file 116/284: water_body_7052.jpg
Processing file 117/284: water_body_7057.jpg
Processing file 118/284: water_body_7067.jpg
Processing file 119/284: water_body_7068.jpg
Processing file 120/284: water_body_7079.jpg
Processing file 121/284: water_body_7100.jpg
Processing file 122/284: water_body_7132.jpg
Processing file 123/284: water_body_714.jpg
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Processing file 124/284: water_body_7166.jpg
Processing file 125/284: water_body_7168.jpg
Processing file 126/284: water_body_7173.jpg
Processing file 127/284: water_body_7181.jpg
Processing file 128/284: water_body_7191.jpg
Processing file 129/284: water_body_7199.jpg
Processing file 130/284: water_body_72.jpg
Processing file 131/284: water_body_7207.jpg
Processing file 132/284: water_body_7235.jpg
Processing file 133/284: water_body_7238.jpg
Processing file 134/284: water_body_7245.jpg
Processing file 135/284: water_body_7247.jpg
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Processing file 137/284: water_body_7265.jpg
Processing file 138/284: water_body_7275.jpg
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Processing file 141/284: water_body_7289.jpg
Processing file 142/284: water_body_7290.jpg
Processing file 143/284: water_body_7296.jpg
Processing file 144/284: water_body_7305.jpg
Processing file 145/284: water_body_7327.jpg
Processing file 146/284: water_body_7386.jpg
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Processing file 150/284: water_body_7401.jpg
Processing file 151/284: water_body_7413.jpg
Processing file 152/284: water_body_7416.jpg
Processing file 153/284: water_body_7417.jpg
Processing file 154/284: water_body_7418.jpg
Processing file 155/284: water_body_742.jpg
Processing file 156/284: water_body_7421.jpg
Processing file 157/284: water_body_7423.jpg
Processing file 158/284: water_body_7433.jpg
Processing file 159/284: water_body_7438.jpg
Processing file 160/284: water_body_744.jpg
Processing file 161/284: water_body_7486.jpg
Processing file 162/284: water_body_7499.jpg
Processing file 163/284: water_body_7510.jpg
Processing file 164/284: water_body_7511.jpg
Processing file 165/284: water_body_7543.jpg
Processing file 166/284: water_body_7552.jpg
Processing file 167/284: water_body_7562.jpg
Processing file 168/284: water_body_7575.jpg
Processing file 169/284: water_body_7578.jpg
Processing file 170/284: water_body_7581.jpg
Processing file 171/284: water_body_7586.jpg
Processing file 172/284: water_body_759.jpg
Processing file 173/284: water_body_7603.jpg
Processing file 174/284: water_body_7615.jpg
Processing file 175/284: water_body_7618.jpg
Processing file 176/284: water_body_7629.jpg
Processing file 177/284: water_body_7638.jpg
Processing file 178/284: water_body_7673.jpg
Processing file 179/284: water_body_7701.jpg
Processing file 180/284: water_body_771.jpg
Processing file 181/284: water_body_7712.jpg
Processing file 182/284: water_body_7717.jpg
Processing file 183/284: water_body_7766.jpg
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Processing file 184/284: water_body_7793.jpg
Processing file 185/284: water_body_78.jpg
Processing file 186/284: water_body_7810.jpg
Processing file 187/284: water_body_7843.jpg
Processing file 188/284: water_body_7871.jpg
Processing file 189/284: water_body_788.jpg
Processing file 190/284: water_body_7891.jpg
Processing file 191/284: water_body_790.jpg
Processing file 192/284: water_body_7901.jpg
Processing file 193/284: water_body_7914.jpg
Processing file 194/284: water_body_7918.jpg
Processing file 195/284: water_body_7956.jpg
Processing file 196/284: water_body_796.jpg
Processing file 197/284: water_body_7961.jpg
Processing file 198/284: water_body_7975.jpg
Processing file 199/284: water_body_7977.jpg
Processing file 200/284: water_body_800.jpg
Processing file 201/284: water_body_8005.jpg
Processing file 202/284: water_body_801.jpg
Processing file 203/284: water_body_8018.jpg
Processing file 204/284: water_body_8033.jpg
Processing file 205/284: water_body_8052.jpg
Processing file 206/284: water_body_8064.jpg
Processing file 207/284: water_body_8140.jpg
Processing file 208/284: water_body_8146.jpg
Processing file 209/284: water_body_8156.jpg
Processing file 210/284: water_body_8159.jpg
Processing file 211/284: water_body_8169.jpg
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Processing file 217/284: water_body_8273.jpg
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Processing file 233/284: water_body_8491.jpg
Processing file 234/284: water_body_8495.jpg
Processing file 235/284: water_body_8501.jpg
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Processing file 237/284: water_body_8506.jpg
Processing file 238/284: water_body_8509.jpg
Processing file 239/284: water_body_851.jpg
Processing file 240/284: water_body_8526.jpg
Processing file 241/284: water_body_8533.jpg
Processing file 242/284: water_body_8546.jpg
Processing file 243/284: water_body_8556.jpg
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Processing file 244/284: water_body_8561.jpg
Processing file 245/284: water_body_8573.jpg
Processing file 246/284: water_body_8584.jpg
Processing file 247/284: water_body_8616.jpg
Processing file 248/284: water_body_8632.jpg
Processing file 249/284: water_body_8636.jpg
Processing file 250/284: water_body_8641.jpg
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Processing file 254/284: water_body_8678.jpg
Processing file 255/284: water_body_8679.jpg
Processing file 256/284: water_body_8694.jpg
Processing file 257/284: water_body_8717.jpg
Processing file 258/284: water_body_8728.jpg
Processing file 259/284: water_body_8730.jpg
Processing file 260/284: water_body_8738.jpg
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Processing file 262/284: water_body_8752.jpg
Processing file 263/284: water_body_8753.jpg
Processing file 264/284: water_body_8770.jpg
Processing file 265/284: water_body_8774.jpg
Processing file 266/284: water_body_8787.jpg
Processing file 267/284: water_body_8792.jpg
Processing file 268/284: water_body_881.jpg
Processing file 269/284: water_body_8847.jpg
Processing file 270/284: water_body_8850.jpg
Processing file 271/284: water_body_8851.jpg
Processing file 272/284: water_body_8859.jpg
Processing file 273/284: water_body_889.jpg
Processing file 274/284: water_body_8912.jpg
Processing file 275/284: water_body_8921.jpg
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Processing file 277/284: water_body_8934.jpg
Processing file 278/284: water_body_8935.jpg
Processing file 279/284: water_body_918.jpg
Processing file 280/284: water_body_949.jpg
Processing file 281/284: water_body_96.jpg
Processing file 282/284: water_body_964.jpg
Processing file 283/284: water_body_968.jpg
Processing file 284/284: water_body_994.jpg
Batch processing completed. Results saved to waterbody-dataset\pred_test_frac_100
[eval] matched 284/284 files in waterbody-dataset\pred_test_frac_100
VAL  -> OA:0.8058 | IoU:0.3559 | Dice:0.4184
TEST -> OA:0.7974 | IoU:0.3736 | Dice:0.4387
Model params: total=24,436,514 | trainable=24,436,514 | size=93.38 MB
Train time: 757.0s | Inference: 230.1 ms/img
```

In [23]:

```
df_unet = pd.DataFrame(results).sort_values("fraction_pct").reset_index(drop=Tru
print("\n== Summary ==")
display(df_unet)
```

```
== Summary ==
```

	fraction_pct	n_train_imgs	params_total	params_trainable	model_size_MB	train_time_s
0	10	227	24436514	24436514	93.384	84.280
1	20	455	24436514	24436514	93.384	153.778
2	30	682	24436514	24436514	93.384	206.972
3	50	1136	24436514	24436514	93.384	378.696
4	100	2273	24436514	24436514	93.384	757.001

In [26]:

```
# ----------------------- plots (OA / IoU / Dice) -----------------------
def _ylims(*series, floor=0.0, ceil=1.0, pad=0.05):
    vals = np.concatenate([np.asarray(s, float)[~np.isnan(s)] for s in series if
    if vals.size == 0:
        return (floor, ceil)
    lo, hi = float(vals.min()), float(vals.max())
    span = max(1e-6, hi - lo)
    return (max(floor, lo - pad*span), min(ceil, hi + pad*span))

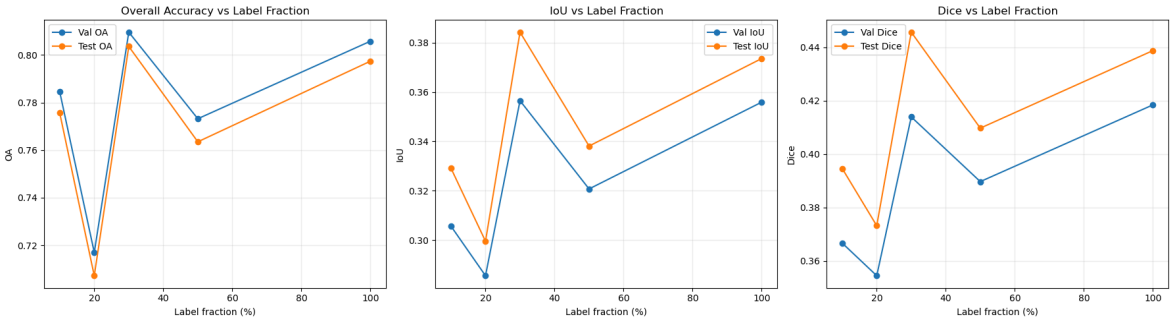
plt.figure(figsize=(18,5))

# OA
ax1 = plt.subplot(1,3,1)
ax1.plot(df_unet["fraction_pct"], df_unet["val_OA"],  "o-", label="Val OA")
ax1.plot(df_unet["fraction_pct"], df_unet["test_OA"], "o-", label="Test OA")
ax1.set_title("Overall Accuracy vs Label Fraction")
ax1.set_xlabel("Label fraction (%)"); ax1.set_ylabel("OA")
ax1.set_ylim(*_ylims(df_unet["val_OA"], df_unet["test_OA"], floor=0.6, ceil=1.0)
ax1.grid(True, alpha=0.25); ax1.legend()

# IoU
ax2 = plt.subplot(1,3,2)
ax2.plot(df_unet["fraction_pct"], df_unet["val_IoU"],  "o-", label="Val IoU")
ax2.plot(df_unet["fraction_pct"], df_unet["test_IoU"], "o-", label="Test IoU")
ax2.set_title("IoU vs Label Fraction")
ax2.set_xlabel("Label fraction (%)"); ax2.set_ylabel("IoU")
ax2.set_ylim(*_ylims(df_unet["val_IoU"], df_unet["test_IoU"], floor=0.2, ceil=1.
ax2.grid(True, alpha=0.25); ax2.legend()

# Dice
ax3 = plt.subplot(1,3,3)
ax3.plot(df_unet["fraction_pct"], df_unet["val_Dice"],  "o-", label="Val Dice")
ax3.plot(df_unet["fraction_pct"], df_unet["test_Dice"], "o-", label="Test Dice")
ax3.set_title("Dice vs Label Fraction")
ax3.set_xlabel("Label fraction (%)"); ax3.set_ylabel("Dice")
ax3.set_ylim(*_ylims(df_unet["val_Dice"], df_unet["test_Dice"], floor=0.2, ceil=
ax3.grid(True, alpha=0.25); ax3.legend()

plt.tight_layout(); plt.show()
```


Overall Accuracy vs Label Fraction
Val OA
Test OA
OA
Label fraction (%)
IoU vs Label Fraction
Val IoU
Test IoU
IoU
Label fraction (%)
Dice vs Label Fraction
Val Dice
Test Dice
Dice
Label fraction (%)

In [ ]: